

In Vivo RNA Interference Screens Identify Regulators of Antiviral CD4⁺ and CD8⁺ T Cell Differentiation

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SUMMARY

Classical genetic approaches to examine the requirements of genes for T cell differentiation during infection are time consuming. Here we developed a pooled approach to screen 30–100+ genes individually in separate antigen-specific T cells during infection using short hairpin RNAs in a microRNA context (shRNAmir). Independent screens using T cell receptor (TCR)-transgenic CD4⁺ and CD8⁺ T cells responding to lymphocytic choriomeningitis virus (LCMV) identified multiple genes that regulated development of follicular helper (Tfh) and T helper 1 (Th1) cells, and short-lived effector and memory precursor cytotoxic T lymphocytes (CTLs). Both screens revealed roles for the positive transcription elongation factor (P-TEFb) component Cyclin T1 (*Ccnt1*). Inhibiting expression of Cyclin T1, or its catalytic partner Cdk9, impaired development of Th1 cells and protective short-lived effector CTL and enhanced Tfh cell and memory precursor CTL formation in vivo. This pooled shRNA screening approach should have utility in numerous immunological studies.

INTRODUCTION

The differentiation of T cells into effector and memory cells is central to adaptive immunity. Transcription factors (TFs) are central regulators of these differentiation processes. Although most current models of T cell differentiation incorporate relatively few regulatory players and rely heavily on the “master regulator” concept, it is abundantly clear that TFs do not act in isolation and that transcription programs that underlie cell differentiation require the concerted actions of multiple factors, including important inducers or repressors of T cell differentiation pathways (Crotty, 2012; Kaech and Cui, 2012; O’Shea and Paul, 2010; Oestreich and Weinmann, 2012; Pipkin and Rao, 2009; Walsh et al., 2002). One large-scale study of the regulation of

gene expression in different cell types and tissues found that a given murine cell type could be distinguished from other cell types by a network of approximately six TF:TF interactions and that these TF networks were conserved in human cell types (Ravasi et al., 2010). Thus, the intersecting expression and actions of multiple TFs appear to determine cell fate and function. Recent work on T helper 17 (Th17) cell differentiation suggests that this model also applies to T cells (Ciofani et al., 2012).

The differentiation of naive CD8⁺ T cells into CTLs is a key process in immunity to viral infections. The differential development of short-lived effector CTLs and precursors to long-lived memory CTLs are considered alternative cellular “fates” (Chang et al., 2007; Joshi et al., 2007), and understanding this process is critical for prevention and treatment of acute and chronic infections (Doering et al., 2012; Haining and Wherry, 2010; Kaech and Cui, 2012). Activated CD4⁺ T cells can differentiate into a range of different functional subsets, including Th1, Th2, Th17, peripheral Treg (pTreg), and follicular helper (Tfh) cells, which each have potent capacities to regulate immune responses and eliminate pathogens. Among CD4⁺ T cells, follicular T helper cells (Tfh) are the specialized providers of help to B cells (Crotty, 2011). T-cell-dependent antibody responses are important for protection against a wide range of pathogens. Our understanding of Tfh cells is still in the early stages, and there is much to be learned about the pathways that control Tfh cell differentiation.

A number of excellent studies have characterized the mRNA expression profiles of CD8⁺ and CD4⁺ T cells isolated ex vivo during the course of antigen-specific responses (Best et al., 2013; Doering et al., 2012; Kaech and Cui, 2012; Kalia et al., 2010; Choi et al., 2013). However, differential mRNA expression studies are likely to overlook a large number of relevant factors responsible for T cell differentiation. For example, of nearly 2,000 predicted conventional DNA-binding transcription factors in the murine genome (Gray et al., 2004), fewer than 15 have validated roles in effector CD8⁺ T cell differentiation (Kaech and Cui, 2012; Pipkin and Rao, 2009). The same limitations probably hold for Tfh cell differentiation and other CD4⁺ T cell differentiation pathways (Crotty, 2012; Oestreich and Weinmann, 2012; Vahedi et al., 2013). Thus, a functional genetic approach in which inhibition of a large number of genes individually, in separate cells in parallel, during T cell differentiation has the potential to rapidly

identify factors comprising the genetic networks underlying T cell function.

To pursue this objective, we have devised an experimental approach that uses retroviral shRNA libraries to diminish the expression of selected gene products one at a time in antigen-specific T cells. Gene function in antiviral responses is then interrogated in pooled screens in mice. We have demonstrated the utility of this approach in two T cell differentiation processes in vivo: CD8⁺ T cell differentiation into cytotoxic T lymphocytes (CTLs) and CD4⁺ T cell differentiation into Tfh and Th1 cells. Here we showed proof of principle that the roles of multiple genes can be interrogated in parallel in T cells during infection and identified previously unappreciated factors that were involved in these differentiation processes. This approach holds promise to substantially accelerate the understanding of T cell differentiation in vivo.

RESULTS

An Optimized Retroviral Vector to Express shRNAs In Vivo

Transduction of activated T cells with murine stem cell virus (MSCV)-based retroviral expression vectors (RVs) has previously been used to drive transgene expression or to deplete expression of endogenous genes by triggering RNA interference (RNAi) using shRNAs upon adoptive transfer in vivo (Araki et al., 2009; Johnston et al., 2009; Joshi et al., 2007; Kao et al., 2011). However, we found that transduction of SMARTA TCR transgenic CD4 T cells (LCMV-specific, gp66-77 IAb restricted) with an MSCV-based (pLMP-derived) RV designed to express shRNAs in the context of miRNA-30 sequences (shRNAmir) resulted in depletion of the transduced cells after an acute LCMV infection (Figure S1A, left, available online). This most likely was due to immune rejection of antigens expressed from pLMP (Figure S1B), because deletion of the puromycin resistance gene from pLMP (LMPd) eliminated this effect (Figure S1A, right). We replaced GFP in LMPd with the violet-excitable, yellow-fluorescing GFP variant Ametrine1.1 (LMP-Amt) to expand its utility in FACS (Figures S1B and S1C) and confirmed its functionality for RNAi in vivo by targeting *Bcl6* transcripts. Transferred SMARTA CD4⁺ T cells transduced with *Bcl6*-specific shRNAs (LMP-Amt sh*Bcl6*-RV, referred to hereafter as sh*Bcl6*-RV) displayed a reduced fraction of CXCR5⁺Bcl6⁺ cells upon LCMV infection, consistent with a requirement for *Bcl6* for differentiation of follicular T helper cells (Tfh) (Figure S1D).

A Pooled Screening System using shRNAs in CD8⁺ T Cells during LCMV Infection

We parallelized the shRNAmir-RV approach in order to interrogate the functions of numerous genes simultaneously. The experimental strategy was to introduce a pool of TCR transgenic T cells carrying individual shRNAs into host mice and assay alterations in the composition of shRNAs carried by the responding T cells during a viral infection (Figure 1A). In effect, each T cell is barcoded by the integrated shRNA-RV, and the fate of individual cells carrying each shRNA can be monitored in T cell populations of interest by deep sequencing DNA libraries derived from the integrated provirus (Figures 1B and 1C; Beronja et al., 2013; Zuber et al., 2011). We optimized conditions in 96-well

format to produce arrays of high-titer RV supernatants without concentration, sufficient to transduce $\geq 70\%$ of LCMV-specific P14 TCR transgenic CD8⁺ T cells 18 hr after TCR stimulation (Figures 1B, S2A, and S2B). The day after transduction, cells from each well were pooled (Figure 1B, day 0) and immediately transferred to recipient mice without cell sorting (sorting reduced P14 accumulation in vivo; Figure S2C), and the recipients were infected with LCMV 1 hr later. In addition, an aliquot of Ametrine-high cells was FACS purified and saved as the “input.”

Genomic DNA was prepared from the input and samples of P14 cells isolated by flow cytometry on day 7 after LCMV infection. Deep sequencing was used to quantify shRNA representation (Figures 1A and S2D–S2H) in libraries generated from a single-step PCR of the shRNAmir sequences in genomic DNA template (Figures 1B and 1C). Multiple PCR conditions were interrogated (Figures S2D–S2G). Independent libraries generated from different DNA template amounts at low PCR cycles (22 or 26 cycles; Figures S2F and S2G) exhibited high correlations in shRNA representation, with both 314 (medium-density) and 318 (high-density) PGM sequencing chips (Figure S2H). Thus, the sequencing approach was robust.

To establish conditions for screening pools of shRNAmir-RV⁺ P14 CD8⁺ T cells in the context of infection, numerous factors were optimized and standardized (Figure S3). Naive Thy1.1⁺ Blimp1-YFP transgenic P14 cells were activated in vitro and transferred to B6 hosts subjected to LCMV infection and the P14 cells were examined as a function of (1) cell transfer number (Figure S3A), (2) the timing of the infection relative to cell transfer (data not shown), (3) LCMV dose (Figure S3B), and (4) LCMV strain (Figure S3C). Transfer of 500,000 activated P14 cells followed by intraperitoneal (i.p.) infection with 1.5×10^5 PFU of LCMV-clone 13 (LCMV-cl13) resulted in a robust infection that induced accumulation of $\sim 10^6$ P14 cells in the spleen by day 7, ~ 50 -fold more than in uninfected recipients (Figures 1D and S3D). Under these conditions virus replication was strongly inhibited (see below), and the responding P14 cells exhibited CD8⁺ T cell phenotypes typical of acute infection, based on interleukin-2 receptor α (IL-2R α) (CD25), KLRG-1, IL-7R α (CD127), and Blimp1-YFP reporter expression (Figures 1E, 1F, and S4A). LCMV-cl13 is more virulent than LCMV Armstrong (Wherry et al., 2003) but was controlled due to the P14 cell transfers. In addition, we confirmed that short-lived effector (KLRG-1^{hi}IL-7R α ^{lo}) and memory precursor (KLRG-1^{lo}IL-7R α ^{hi}) P14 populations exhibited different potentials for memory cell formation and “recall” capacity (Figures S4B–S5E). Altogether, these results demonstrate robust conditions with which to screen effector and memory CTL development.

The number of distinct shRNAmirs that could be tested in parallel was constrained by the number of adoptively transferred T cells. In order to ensure library complexity, we aimed to represent each shRNA with ~ 500 cells per mouse upon engraftment. This depth of representation is similar to or exceeds recent in vivo shRNA-based screens (Beronja et al., 2013; Zhou et al., 2014; Zuber et al., 2011). Based on data from adoptively transferred naive CD8⁺ T cells (Badovinac et al., 2007) and activated CD8⁺ T cells (Pipkin et al., 2010), we assumed that $\sim 10\%$ of transferred cells would engraft. Thus, we initially analyzed a pool of 500,000 cells representing ~ 100 unique shRNAs in a single experiment. We also considered the recovery of effector and

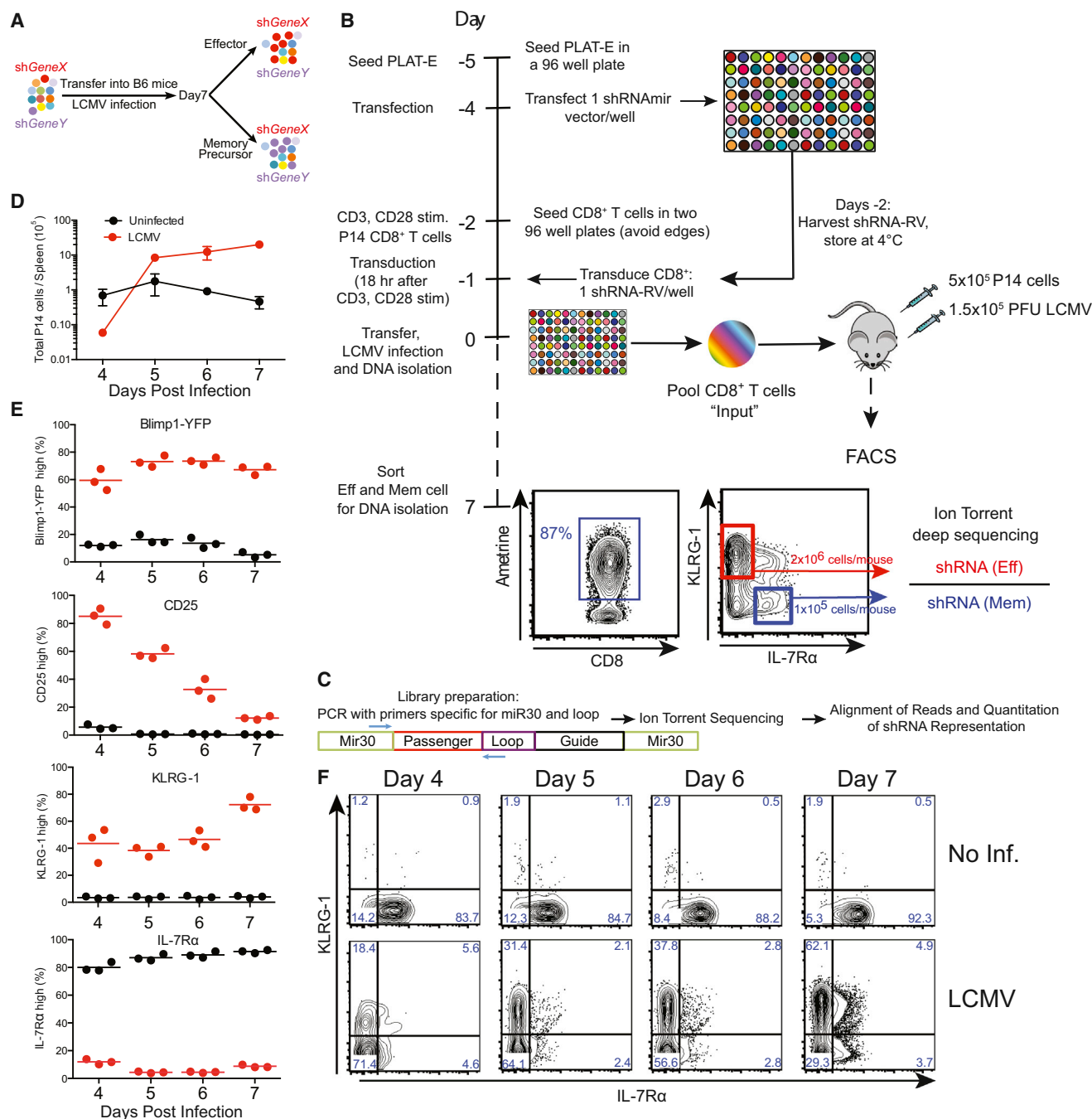


Figure 1. Optimization of Conditions for a Pooled Screening Approach using shRNAmirs in CD8⁺ T Cells In Vivo to Identify Genes that Regulate CTL Differentiation during Infection

(A) A conceptual representation depicting the principle of the pooled screening strategy.

(B) Scheme for the shRNAmir screen using P14 cells and LCMV infection.

(C) Scheme for quantifying shRNAmirs. DNA libraries generated by PCR of the integrated shRNAmir provirus are analyzed by deep sequencing to quantify shRNA representation in the cell subsets.

(D) Total P14 cell numbers recovered in the spleen in the presence or absence of LCMV infection. Error bars indicate standard deviations.

(E) Blimp1-YFP^{hi}, CD25^{hi}, KLRG-1^{hi}, and IL-7Rα^{hi} cell frequencies at the indicated time points after infection. Symbols represent values from individual mice. Red indicates LCMV-infected mice; black indicates uninfected mice.

(F) Representative flow cytometry plots of KLRG-1 and IL-7Rα staining on P14 cells under conditions used for screening.

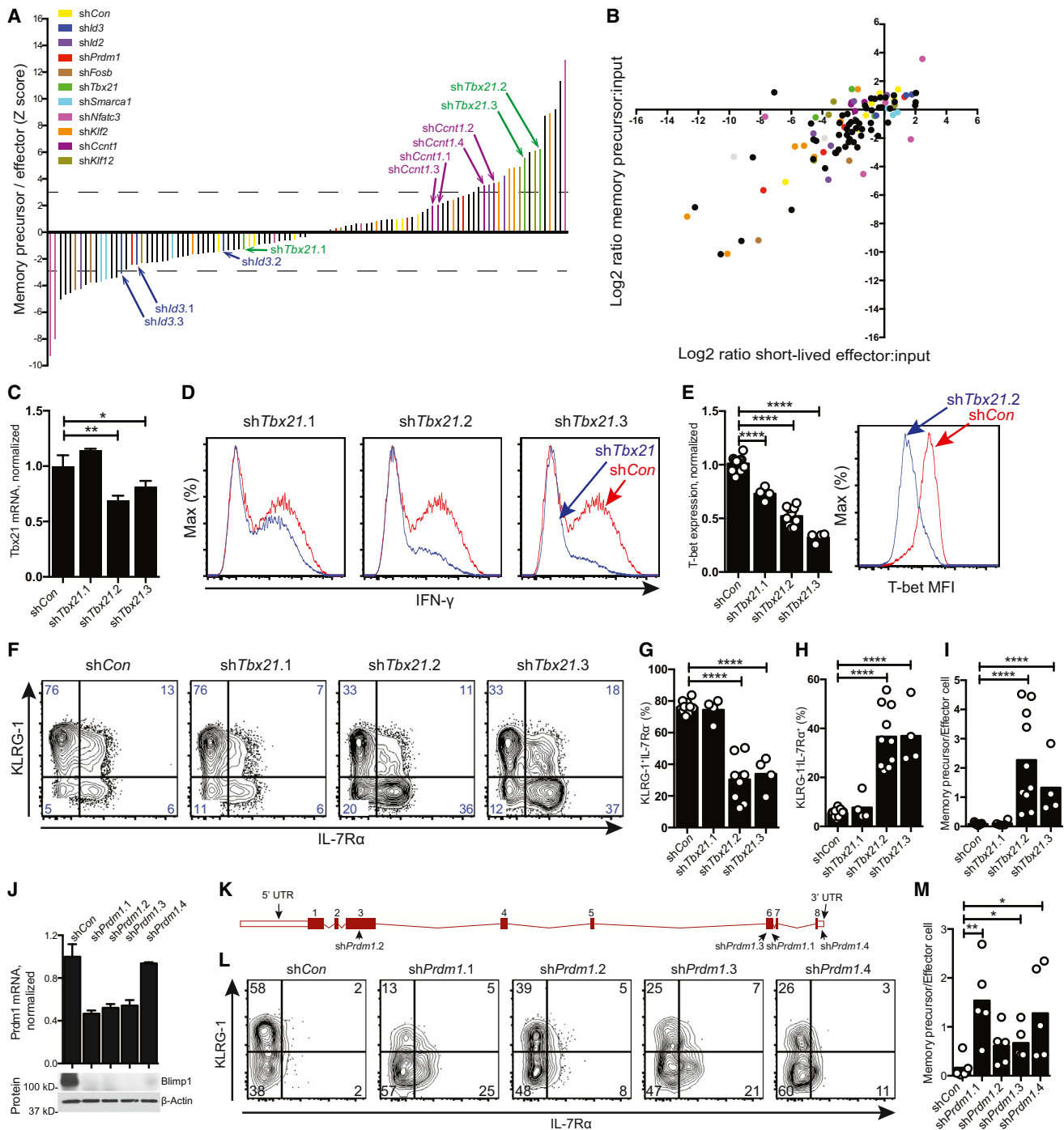


Figure 2. A Pooled RNAi Screen in CD8⁺ T Cells In Vivo Identifies Potential Regulators of Effector and Memory Precursor CTL Formation

(A) Relative enrichment of shRNAs in memory precursor and short-lived effector P14 cell populations is reported as Z-scores for each shRNA in the library. Each bar represents a single shRNA. Negative control shRNAs are colored yellow.

(B) Scatter plot shows the log₂ ratio of normalized reads of all shRNAs in each sorted CD8⁺ T cell subset versus the input sample. Each dot represents a unique shRNA and is color coded as in (A).

(C) *Tbx21* mRNA expression in shTbx21⁺ P14 CD8⁺ T cells, after 6 days of culture (10 U/ml IL-2). Abbreviation is as follows: shCon, control shRNAmir.

(D) Intracellular IFN- γ staining in P14 CD8⁺ T cells, gated on shTbx21⁺ cells. Cells were cultured for 6 days (10 U/ml IL-2) and restimulated with PMA and ionomycin for 4 hr before staining.

(E) T-bet expression in shTbx21⁺ P14 CD8⁺ T cells from spleens 8 days after LCMV infection (normalized geometric MFI). T-bet staining is shown for representative mice (right).

(F) Contour plots show KLRG-1 and IL-7R α staining on shTbx21⁺ P14 CD8⁺ T cells from representative mice 8 days after LCMV infection.

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memory precursor populations. Based on P14 accumulation data at day 7 postinfection (Figure 1D), we expected to recover ~300,000 KLRG-1^{hi}IL-7R α ^{lo} and ~30,000 KLRG-1^{lo}IL-7R α ^{hi} cells per mouse. To assure that cell numbers would not limit library complexity at the end of the experiment and to reduce potential founder effects from variation in individual mice (Zuber et al., 2011), sorted cells were pooled from five or more infected mice in each experiment.

Identification of Genes Underlying CTL Differentiation by an shRNA Screen In Vivo

We selected 34 genes to screen to test the approach. These included genes differentially expressed in Tfh cells and CTLs (Choi et al., 2013), broadly expressed chromatin and transcriptional regulators with unknown roles in effector T cells, and multiple positive and negative control genes. Each gene was targeted by 1–5 shRNAmirs, depending on availability in the original library (see Experimental Procedures and Table S1). The use of multiple shRNAs per gene is important, to control for off-target effects, and not all shRNAs are functional. A total of 110 unique shRNAs that target these genes were subcloned as a pool into pLMPd-Amt (Table S1 and Figures S5A and S5B). Individual colonies from this transformation were picked, sequence verified, and rearranged into 96-well plates.

Retroviral supernatants prepared from cells transfected with the array of 110 shRNAmir DNAs were used to transduce P14 cells. Transduction of each construct was confirmed by Ametrine fluorescence, and P14 cells were pooled. An aliquot was removed for the input sample, and 500,000 cells were transferred into multiple recipient mice that were then infected with LCMV. On day 7 postinfection, short-lived effector (KLRG-1^{hi}IL-7R α ^{lo}) and memory precursor (KLRG-1^{lo}IL-7R α ^{hi}) P14 cells were sorted, genomic DNA was extracted from each population, and shRNAmir sequencing libraries were prepared. After sequencing and alignment to the reference shRNA sequences, 642,718 (input), 233,902 (effector), and 487,865 (memory) mapped reads were retained for each cell subset. All intended shRNAs introduced during T cell transduction were recovered from input samples; only two were not detected in either the effector or memory precursor cell populations. The memory precursor:short-lived effector cell ratio for each shRNA was calculated. Values from negative control shRNAs targeting genes not expressed in CD8⁺ T cells (*Cd4*, *Cd14*, *Cd19*, *Ms4a1* [CD20]) based on RNA-seq analysis (M.E.P., unpublished data) were used to calculate Z-scores for each shRNA (Figure 2A and Table S1) and the ratios of shRNAs in effector and memory precursor subsets relative to the input cells was plotted (Figure 2B). Several shRNAs were substantially reduced

in both short-lived effector and memory precursor P14 cell subsets relative to input, suggesting that genes they affected were required for the accumulation of P14 T cells during infection (Figure 2B).

To focus on factors with differential effects on short-lived effector versus memory precursor CD8⁺ T cell subsets, we identified genes for which two or more cognate shRNAs were enriched with Z-score values of $\geq |3.0|$ and classified these as hits. None of the negative control genes exhibited this pattern (Figure 2A and Table S1). Genes that met these criteria were identified in both effector and memory precursor subsets (Figure 2A and Table S1). As expected from studies with gene-deficient mice (Cannarile et al., 2006; Intlekofer et al., 2005; Joshi et al., 2007; Yang et al., 2011), *Tbx21* (T-bet)- and *Id2*-specific shRNAs were enriched in memory precursor cells, because these genes are necessary for effector CTL generation (Figure 2A and Table S1). Conversely, all three *Id3*-specific shRNAs were enriched in effector CTLs (Figure 2A and Table S1), but just below the criteria to be designated a hit, consistent with a mild early defect in memory precursor formation in *Id3*-deficient mice (Ji et al., 2011; Yang et al., 2011).

Tbx21 and *Prdm1* shRNAs Impair Effector CTL Development during LCMV Infection

We validated results of the screen by examining the impact of shRNAs individually. Both sh*Tbx21.2* and sh*Tbx21.3*, but not sh*Tbx21.1*, strongly depleted *Tbx21* mRNA and T-bet protein (Figures 2C and 2E), inhibited interferon- γ (IFN- γ) expression (Figure 2D), and limited development of short-lived effector cells in vivo ($p < 0.01$, Figures 2F–2I). These results correlated directly with the enrichment of these shRNAs in the screen (Figure 2A and Table S1) and confirmed the role of T-bet in the generation of effector CD8⁺ T cells.

Prdm1 (encoding Blimp-1) has known roles in effector CD8⁺ and CD4⁺ T cell differentiation (Rutishauser et al., 2009; Shin et al., 2009; Johnston et al., 2009). Consistent with this, *Prdm1* shRNAs were preferentially enriched in memory precursor CD8⁺ T cells, but the magnitude of their effects differed in replicates of the in vivo screen (Figure S5C and Table S1). Analysis of each *Prdm1* shRNA individually showed that three of four shRNAs impaired expression of both Blimp-1 mRNA and protein (Figure 2J). The fourth shRNA impaired Blimp-1 protein expression but did not reduce its mRNA (Figure 2J), perhaps because it targeted the *Prdm1* 3' UTR (Figure 2K). Individually, all four *Prdm1* shRNAs impaired effector CD8⁺ T cell frequencies and increased the ratio of memory precursor cells to short-lived effector cells in vivo ($p < 0.01$ – 0.05 , Figures 2L and 2M). These data indicate that shRNAs can have variable effects but

(G–I) Quantitation of CD8⁺ T cell subsets resulting from sh*Tbx21*⁺ P14 cells in vivo.

(G) Short-lived effector cells (KLRG-1^{hi}IL-7R α ^{lo}).

(H) Memory precursor cells (KLRG-1^{lo}IL-7R α ^{hi}).

(I) Ratio of memory precursor to short-lived effector phenotype P14 cells, per mouse.

(J) *Prdm1* mRNA expression was determined by qRT-PCR in transduced P14 CD8⁺ T cells after sorting from spleens 7 days after LCMV infection. Blimp1 protein expression was determined by immunoblot analysis after 4 days of culture with IL-12 (5 ng/ml) and IL-2 (100 U/ml).

(K) Map of *Prdm1* with shRNA-targeted regions indicated.

(L) Contour plots of KLRG-1 and IL-7R α staining on sh*Prdm1*⁺ P14 CD8⁺ T cells from representative mice at 7 days after LCMV infection.

(M) Ratios of memory precursor to effector P14 CD8⁺ T cells. Each symbol represents T cells from an individual mouse.

Data are pooled from three (H, I) and two (J, L) independent experiments. * $p < 0.05$, ** $p < 0.01$, **** $p < 0.0001$. Error bars represent standard deviations.

confirmed that *Prdm1* expression is required for short-lived effector CD8⁺ T cell differentiation.

Identification of Genes Underlying Tfh Cell Differentiation via an shRNA Screen In Vivo

In parallel, we developed a pooled screen in CD4⁺ T cells to discover genes important for Tfh and Th1 cell differentiation in vivo (Figure 3A). A total of 5×10^5 shRNAmir⁺Amt^{hi} SMARTA cells were transferred into B6 hosts, and mice were infected 3–4 days later with LCMV Armstrong (Figure 3A). DNA was also isolated from an aliquot of cells before the transfer (input). At 6 days after LCMV infection, virus-specific Tfh cells (CXCR5⁺SLAMF^{lo}) and Th1 cells (CXCR5⁺SLAMF^{hi}) (Choi et al., 2013; Johnston et al., 2009) were isolated by flow cytometry and deep sequenced for differential representation of the shRNAs. The Tfh:Th1 ratios for each shRNA were calculated and their Z-scores were plotted (Figures 3B and 3C). A total of 14 control shRNAs expected not to affect Tfh or Th1 cell differentiation (not known to be expressed in CD4⁺ T cells: *Cd14*, *Cd19*, *Cd22*, *Ms4a1*, *Cd8*, *Smarca1*) were equally distributed in both populations (Figure 3B), and their effects on cell accumulation were also assessed (Figure 3C). Based on a Z-score cutoff of $\geq |3.0|$ for each shRNA, factors encoded by *Prdm1*, *Chd4*, *Id2*, and *Ccnt1* were identified as candidate positive regulators of Th1 cells or inhibitors of Tfh cell differentiation (Figures 3B and 3C, Table S1). *Prdm1* is a positive control, as it encodes an inhibitor of Tfh cell differentiation (Johnston et al., 2009), and is discussed further below. Genes *Fosb*, *Plagl1*, *Mta3*, and *Runx3* were identified as potential positive regulators of Tfh cells or inhibitors of Th1 cell differentiation (Table S1). *Plagl1* is highly expressed in Tfh cells (Hale et al., 2013; Yusuf et al., 2010), and MTA3 (encoded by *Mta3*) is known to interact with Bcl6 in B cells (Fujita et al., 2004). Itch is a known positive regulator of Tfh cell differentiation, based on a profound loss of Tfh cells in *Itch*^{fl/fl}*Cd4-cre*⁺ mice (Xiao et al., 2014). Notably, all four *Itch* shRNAs were severely depleted from Tfh cells and highly enriched in the Th1 cell population (Figures 3B and 3C). In a second replicate of the CD4⁺ T cell screen, the validated *Bcl6* shRNA (sh*Bcl6*.2, Figure S1D) was depleted from the Tfh cell population, as expected (Figure 3D). Comparisons of the two independent screens indicated that the in vivo screens generated reproducible results (Figure 3E) and also showed that shRNA representation was similar even when the libraries were sequenced with increased coverage using the higher-density PGM 318 chip (Figure 3E). These results suggest that the CD4⁺ T cell shRNAmir-RV screening approach in vivo was also robust.

To confirm results from the primary screen (Figure 3B), the effects of *Prdm1* shRNAs were examined individually. SMARTA CD4⁺ T cells transduced with sh*Prdm1*.1-RV exhibited the strongest Tfh cell bias in vivo ($p < 0.001$, Figures 3F and 3G), consistent with results from the screen. A modest but significant Tfh cell bias was observed in sh*Prdm1*.3⁺ and sh*Prdm1*.4⁺ SMARTA CD4⁺ T cells when compared to untransduced CD4⁺ T cells ($p < 0.01$, Figures 3F and 3G). sh*Prdm1*.2 had no effect on Tfh cell differentiation (Figures 3F and 3G), consistent with it having the weakest effect on *Prdm1* mRNA expression in CD4⁺ T cells (data not shown). These results correlated with the observed distribution of the four shRNAs in the primary screen in CD4⁺ T cells (Figure 3B) and also indicate that the activity of individual

shRNAs might depend on the specific cellular context (e.g., CD4⁺ versus CD8⁺ T cells; Figures 3B and 2G). Thus, results of the pooled shRNA screen were consistent with experiments using individual constructs.

Ccnt1 Is Required for Both Th1 Cell and Effector CD8⁺ T Cell Differentiation In Vivo

We compared the full data sets from the CD8⁺ and CD4⁺ T cells screens and found that inhibition of several different genes affected differentiation of both effector CD8⁺ and CD4⁺ T cells (Figures 4A and 4B). *Ccnt1* encodes Cyclin T1, a noncanonical cyclin that is a regulatory subunit of the RNA polymerase II-positive transcription elongation factor (P-TEFb). All four *Ccnt1*-specific shRNAs were depleted from KLRG-1^{hi}IL-7R α ^{lo} effector CD8⁺ T cells and from Th1 cells in the screens (Figure 4A). Based on the notion that functional parallels might exist between differentiation of CD4⁺ and CD8⁺ T cells during infection (Choi et al., 2013; Yang et al., 2011), we further explored the roles of Cyclin T1 in both subsets.

In CD4⁺ T cells, all four *Ccnt1*-shRNAs inhibited Cyclin T1 protein expression to varying degrees; three of four caused robust inhibition (Figures 4C and S6A). Each *Ccnt1* shRNA was examined individually in SMARTA CD4⁺ T cells 6 days after LCMV infection (Figures 4D–4G). Neither CD4⁺ T cell proliferation (Figures S6B and S6C) nor CD4 or CD44 expression was affected by *Ccnt1* shRNAs (data not shown). *Ccnt1* shRNAs both increased Tfh cell development (CXCR5⁺SLAMF^{lo}) and decreased Th1 cell formation ($p < 0.0001$ – 0.05 ; Figures 4D and 4E). Germinal center Tfh (GC Tfh) cells are a fully polarized subset of Tfh cells (CXCR5⁺PSGL1^{lo}; Crotty, 2011; Poholek et al., 2010) and their frequencies were increased by *Ccnt1* shRNAs ($p < 0.001$ – 0.05 ; Figures 4F and 4G).

We also examined T cell differentiation at earlier time points and found that *Ccnt1* shRNAs substantially increased the proportion of early CXCR5⁺Bcl6⁺ Tfh cells ($p < 0.01$; Figures 4H and 4I, and $p < 0.01$ – 0.05 ; Figure S6D). The increased Tfh cell differentiation of sh*Ccnt1*⁺CD4⁺ T cells after LCMV infection could be a reflection of a decreased potential of these cells to differentiate into Th1 cells. Consistent with this hypothesis, *Ccnt1* shRNAs resulted in decreased expression of T-bet in vivo ($p < 0.001$ – 0.05 ; Figure 4J). Reciprocally, the expression of CD40L, an essential component of T cell help to B cells, was also increased in sh*Ccnt1*⁺ cells ($p < 0.001$ – 0.05 ; Figure 4K). These results suggest that Cyclin T1 promotes Th1 cell differentiation at the expense of Tfh cell differentiation in vivo.

To test the requirement for Cyclin T1 in T cell differentiation in vitro, we cultured shCon⁺ and sh*Ccnt1*⁺ CD4⁺ T cells in Th1-cell-biasing conditions. *Ccnt1* shRNAs impaired T-bet expression under these conditions ($p < 0.0001$; Figure 5A), resulting in a substantial loss of IFN- γ production upon restimulation ($p < 0.0001$ – 0.001 ; Figures 5B and 5C). The defect was cell intrinsic, because no defect in IFN- γ production was observed in untransduced CD4⁺ T cells in the same wells (Figure S6E). These results support a model in which reduced Cyclin T1 expression impairs Th1 cell development and favors Tfh cell development.

The P-TEFb Subunit Cdk9 Is Necessary for Th1 Cell Differentiation

Conventional P-TEFb comprises Cdk9 (catalytic subunit) and a regulatory subunit (e.g., Cyclin T1 or T2). To test whether Cyclin

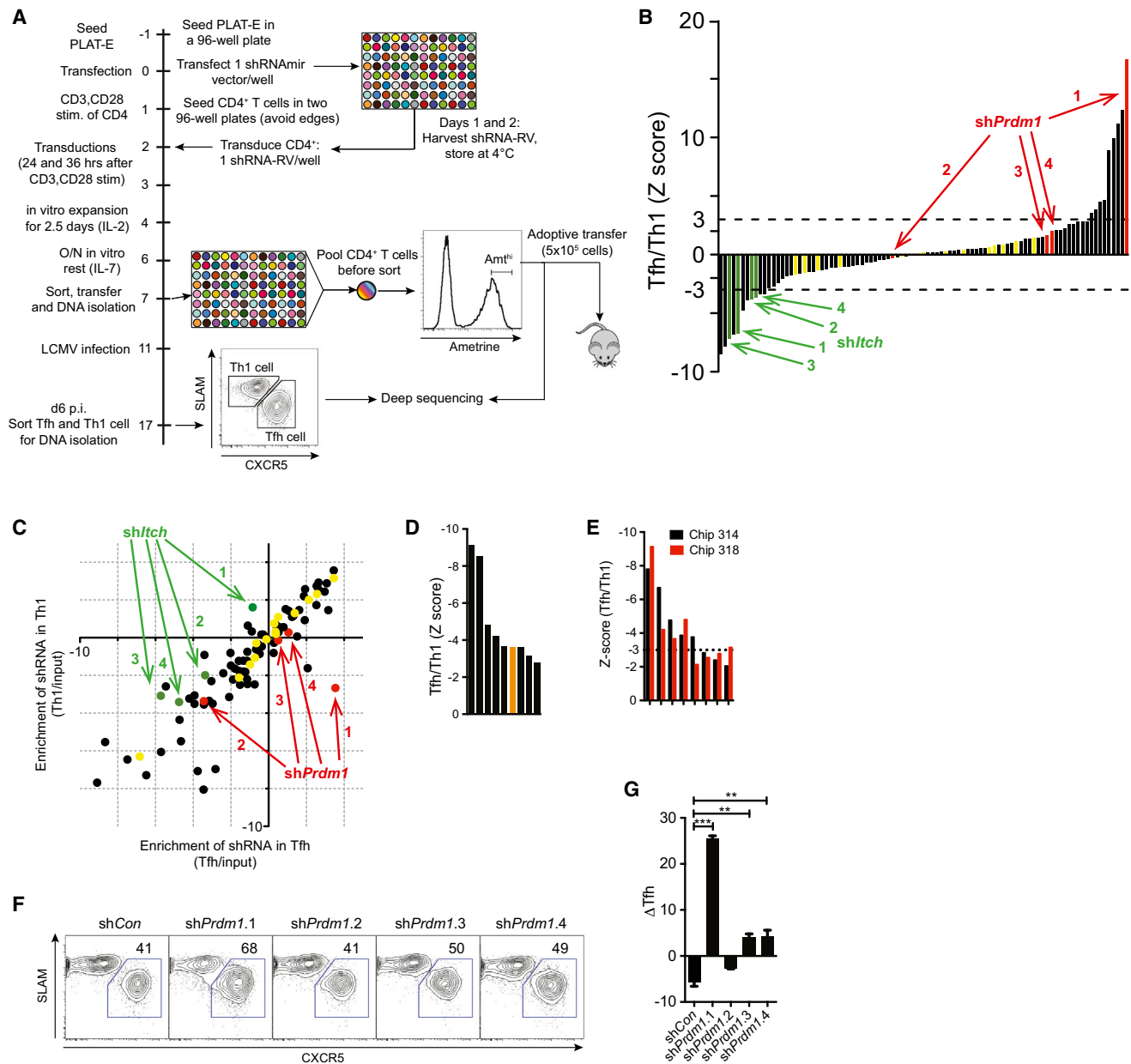


Figure 3. A Pooled RNAi Screen in CD4⁺ T Cells Identifies Potential Regulators of Tfh and Th1 Cell Differentiation In Vivo

(A) Scheme for the shRNA screening approach using SMARTA CD4⁺ T cells.

(B) Relative enrichment of shRNAs in the Tfh or Th1 cell populations in vivo, reported as Z-score values for each shRNA in the library. Z-scores of |3| and |2| are indicated by a dotted line and a tick, respectively.

(C) Scatter plot shows the log₂ of normalized reads of all shRNA in Tfh and Th1 cell populations versus the input sample. This reveals effects on cell survival or proliferation. shRNA are color coded as in (B).

(D) Z-scores are shown for the shRNAs most depleted from the Tfh cell population in the Ion 318 Chip experiment. shBcl6 is highlighted in orange.

(E) Z-scores of shRNAs depleted from Tfh cells in two independent deep sequencing reactions: Ion 314 Chip (black bars) and Ion 318 Chip (red bars). The dotted line is a Z-score of -3.

(F) SMARTA CD4⁺ T cells were transduced with the indicated shRNAs, transferred into B6 mice, and analyzed 6 days after LCMV infection. shCon indicates a control shRNA. Representative flow cytometry plots are shown of shRNA⁺ SMARTA CD4⁺ T cells with Tfh cell (CXCR5⁺SLAMF6⁺) gate drawn.

(G) The differences in percentages of Tfh (%Tfh of Amt⁺ – %Tfh of Amt⁻) for each shRNA in SMARTA CD4⁺ T cells are shown. **p < 0.01, ***p < 0.001. Error bars represent standard deviations.

T1 was likely acting via P-TEFb, we examined two *Cdk9* shRNAs in SMARTA CD4⁺ T cells for their effects on Tfh versus Th1 cell differentiation. Both shRNAs inhibited *Cdk9* expression

in vitro (Figure 5D). CD4⁺ T cells transduced with *Cdk9* shRNAs and cultured under Th1-cell-biasing conditions showed impaired production of IFN-γ, similar to the effect of *Ccnt1*

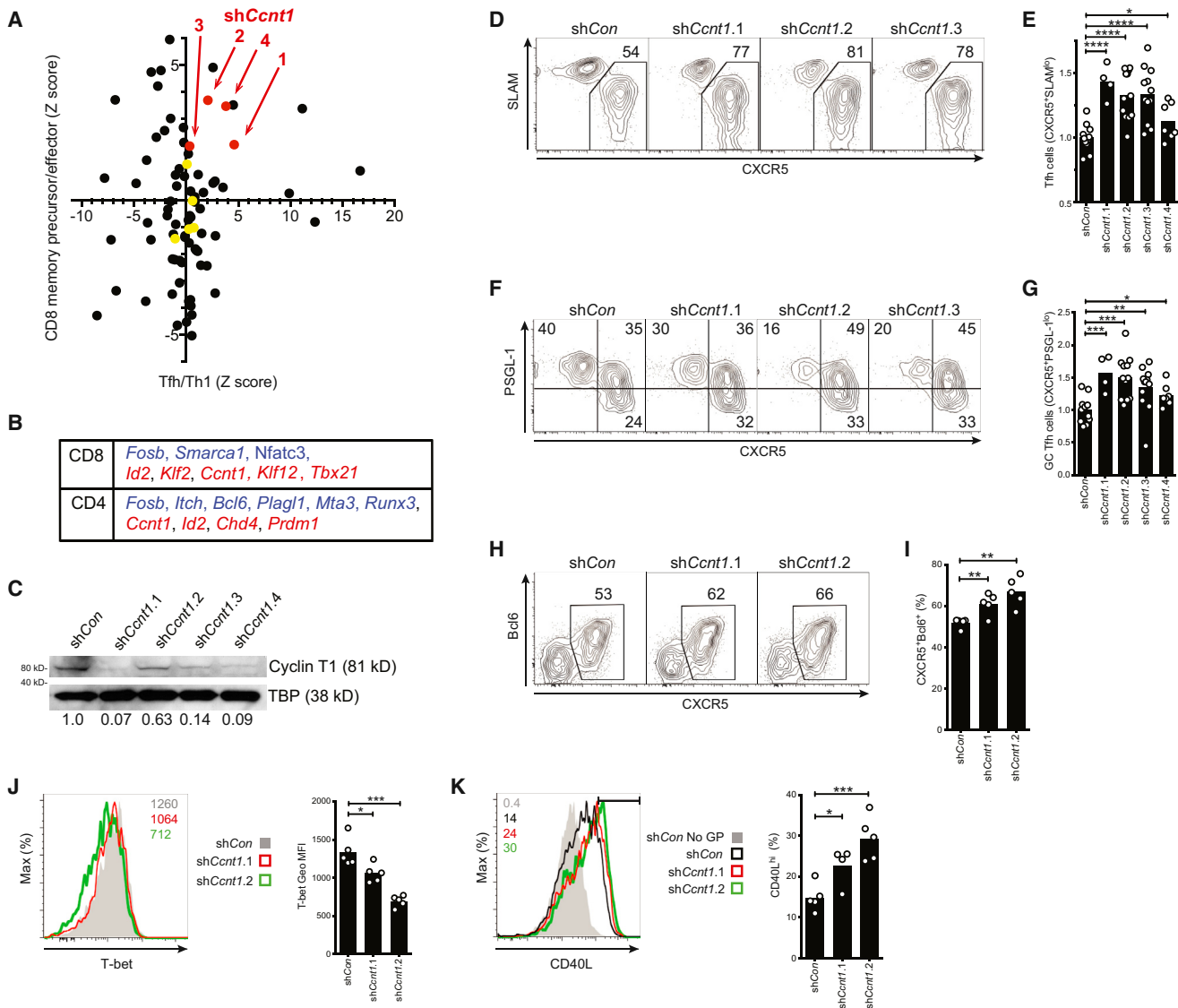


Figure 4. *Ccnt1* Depletion Promotes Development of Tfh Cells during Viral Infection

(A) Comparison of shRNA screening results in both CD4⁺ and CD8⁺ T cells. Tfh and Th1 CD4⁺ T cells differentiation results, plotted against memory precursor and effector CD8⁺ T cell differentiation results. Z-score values are shown. Common negative control shRNAs are shown in yellow, and those targeting *Ccnt1* are highlighted (red).

(B) Table of top hits for genes required for memory precursor CD8⁺ T cell or Tfh CD4⁺ T cell differentiation (blue) and short-lived effector CD8⁺ T cell or Th1 CD4⁺ T cell differentiation (red).

(C) Cyclin T1 protein expression in MCC T cells after transduction with the indicated shRNAs and 4 days of culture. The ratios of Cyclin T1 to TBP relative to the control shRNA are indicated.

(D and E) Flow cytometry plots (D) and quantitation (normalized) (E) of Tfh cell differentiation (CXCR5⁺SLAMF6^{hi}) by shCcnt1⁺ SMARTA CD4⁺ T cells at 6 days after LCMV infection.

(F and G) Flow cytometry plots (F) and quantitation (normalized) (G) of GC Tfh cell differentiation (CXCR5⁺PSGL1^{hi}) by shCcnt1⁺ SMARTA CD4⁺ T cells at 6 days after LCMV infection.

(H and I) Flow cytometry plots (H) and quantitation (I) of CXCR5 and Bcl6 expression by shCcnt1⁺ SMARTA CD4⁺ T cells at 4 days after LCMV infection.

(J) Tbet expression in shCcnt1⁺ SMARTA CD4⁺ T cells in vivo, 3 days after LCMV infection. Tbet geometric MFIs are graphed (right).

(K) CD40L expression by shCcnt1⁺ SMARTA CD4⁺ T cells at 4 days after LCMV infection, after 2 hr restimulation with GP₆₁₋₈₀ peptide. The percentages of CD40L^{hi} cells are indicated. Each symbol represents T cells from an individual mouse.

Data are pooled from three (F, G) or representative of two (K) or three (H–J) independent experiments. *p < 0.05, **p < 0.01, ***p < 0.001, ****p < 0.0001.

shRNAs (p < 0.001–0.01; Figures 5E and 5F), suggesting that both Cyclin T1 and Cdk9 promote Th1 cell differentiation in vitro.

Cdk9 depletion in vivo favored Tfh (CXCR5⁺SLAMF6^{hi}) cell development while reducing Th1 cell differentiation (shCdk9.1, p < 0.0001; Figures 5G and 5H), without impairing T cell expansion

(Figure S7A) or normal expression of CD4 and CD44. Furthermore, Cdk9 depletion increased the frequency of GC Tfh cells, measured as CXCR5⁺PSGL1^{lo} cells (sh*Cdk9.1*, $p < 0.001$; Figures 5I and 5J) or CXCR5⁺Bcl6^{hi} cells (Figures S7B and S7C). Additionally, based on CXCR5 and Bcl6 expression, early Tfh cell differentiation in sh*Cdk9*⁺CD4⁺ T cells was enhanced 4 days after LCMV infection ($p < 0.01$ – 0.05 ; Figures 5K and 5L), with normal cell expansion (Figure S7D). Intriguingly, similar to what was observed in the absence of Cyclin T1, inhibition of *Cdk9* expression also increased CD40L expression by CD4⁺ T cells ($p < 0.001$ – 0.01 ; Figure 5M). Altogether, these data recapitulated those obtained with sh*Ccnt1*⁺CD4⁺ T cells and suggest that P-TEFb might preferentially promote Th1 cell differentiation of activated CD4⁺ T cells.

***Ccnt1* and *Cdk9* Are Required for Development of Protective Effector CTLs In Vivo**

Next, we explored the requirements of *Ccnt1* expression in CD8⁺ T cells. Three of the four *Ccnt1*-specific shRNAs resulted in near complete inhibition of Cyclin T1 protein expression in cultured CD8⁺ T cells; one shRNA exhibited strong but incomplete inhibition (Figure 6A). The accumulation of P14 CD8⁺ T cells transduced with *Ccnt1* shRNAs was not impaired during culture (Figure 6B) or in vivo (Figure 6C). However, *Ccnt1* shRNAs strongly impaired generation of short-lived effector P14 T cells (Figures 6D and 6E). There was a concomitant increase in the fraction of memory precursor phenotype P14 cells (Figures 6D and 6F), which increased the ratio of memory precursor to short-lived effector P14 cells (Figure 6G).

To examine whether the effects of *Ccnt1* shRNAs in CD8⁺ T cells were related to Cyclin T1 function as a subunit of P-TEFb, we examined the effects of *Cdk9*-specific shRNAs (Figure 6H). Similar to sh*Ccnt1*, sh*Cdk9*⁺ P14 cells exhibited reduced short-lived effector cell and increased memory precursor cell formation in vivo (Figures 6I–6K). This correlated with decreased T-bet expression in vivo (Figure 6L). Notably, T-bet expression in sh*Ccnt1*⁺ or sh*Cdk9*⁺ P14 cells was impaired in KLRG-1^{lo}IL-7R α ^{lo} cells (data not shown), a stage that presumably precedes development of either short-lived effector or memory precursor CD8⁺ T cells. These data show that wild-type amounts of Cyclin T1 and Cdk9 are necessary for efficient development of short-lived effector CTLs and that they normally limit generation of memory precursor CD8⁺ T cells.

The defect in sh*Ccnt1*⁺ and sh*Cdk9*⁺ short-lived effector CD8⁺ T cell formation brought into question whether these cells were effective at viral control. Thus, we examined LCMV burden on day 8 of infection. LCMV titers in the spleen of host mice with sh*Ccnt1*⁺ or sh*Cdk9*⁺ P14 cells were at least 2- to 5-fold higher than controls (Figure 7A). Correlating with this finding, sh*Ccnt1*⁺ or sh*Cdk9*⁺ P14 cells from infected mice expressed less granzyme B (Figure 7B). Finally, we examined sh*Ccnt1*⁺ or sh*Cdk9*⁺ P14 cells under culture conditions that strongly induce CTL differentiation (Pipkin et al., 2010). Under these conditions, all shRNAs targeting either *Ccnt1* or *Cdk9* also specifically inhibited perforin protein expression (Figure 7C). Thus, normal amounts of Cyclin T1 and Cdk9 appear to be required for the upregulation of genes encoding cytotoxic effector functions and for effective CTL-mediated protection from viral infection.

DISCUSSION

We have demonstrated the applicability of a pooled approach using shRNAmirs in T cells to screen for factors that regulate CD4⁺ and CD8⁺ T cell differentiation in response to viral infection. Using an improved shRNAmir vector that enhances suitability of shRNA-mediated RNAi in studies that depend upon cell transfers, more than 100 unique shRNAs were screened simultaneously in vivo. In separate proof-of-principle screens during LCMV infection using virus-specific TCR transgenic SMARTA CD4⁺ T cells or P14 CD8⁺ T cells, we identified multiple candidate genes with potential roles in the development of Th1 and Tfh CD4⁺ T cells as well as short-lived effector and memory precursor CD8⁺ T cells. Detailed follow-up analyses of one factor identified in both screens revealed specific roles for Cyclin T1 and Cdk9, components of P-TEFb, for promoting Th1 cell development of CD4⁺ T cells and protective effector CTL development of CD8⁺ T cells during infection, while limiting differentiation of Tfh cells and memory precursor CD8⁺ T cells. These data suggest that regulation of transcription elongation by P-TEFb might be an important mechanism underpinning differentiation of T cells during immune responses.

Several important aspects distinguish the screen presented here from other recently reported pooled in vivo shRNA-screens (Beronja et al., 2013; Zhou et al., 2014; Zuber et al., 2011). The screening approach here used an array strategy to ensure independent transductions and equal representation of shRNAmirs in the input pools. In addition, in contrast to other screens based on shRNA-dependent cell accumulation or depletion as the main readout to identify primary hits (Beronja et al., 2013; Zhou et al., 2014; Zuber et al., 2011), the approach presented here was a phenotypic screen of cell differentiation during an infection. As such, it was complicated by the nature of T cell responses during infection, which are constrained by factors such as the frequencies of antigen-specific T cells (Badovinac et al., 2007; Obar et al., 2008), and thus, is distinct from a T cell adoptive immunotherapy setting, which affords transferring much larger T cell numbers (Zhou et al., 2014).

The ability to conduct large-scale pooled screens using shRNAs has advanced, although interpreting the results of shRNA-based assays remains complicated. As our data on the *Prdm1*, *Tbx21*, and *Ccnt1* genes emphasize, interrogating the effects of multiple shRNAs targeting the same gene in several assays tends to clarify the role of each gene, because each shRNA can result in nonidentical phenotypes attributable to differential effects on particular target RNA isoforms, unintended off-target effects, or partial attenuation of target-specific gene expression. In our experience, ~60% of shRNA sequences derived from the GIPZ library (the source of most shRNAs for this study) impaired target gene expression or caused a measurable biological phenotype (data not shown). However, newer algorithms trained on functional data have improved predicting shRNAmirs that trigger RNAi more potently and specifically than previous designs (Fellmann et al., 2011). Our study employed some of these designs and they are likely to enhance the fidelity of future large-scale screens in vivo.

Using a conservative approach, we showed that more than 100 unique shRNAmirs represented by 500,000 adoptively transferred P14 cells could be assayed in fewer than 10 mice in

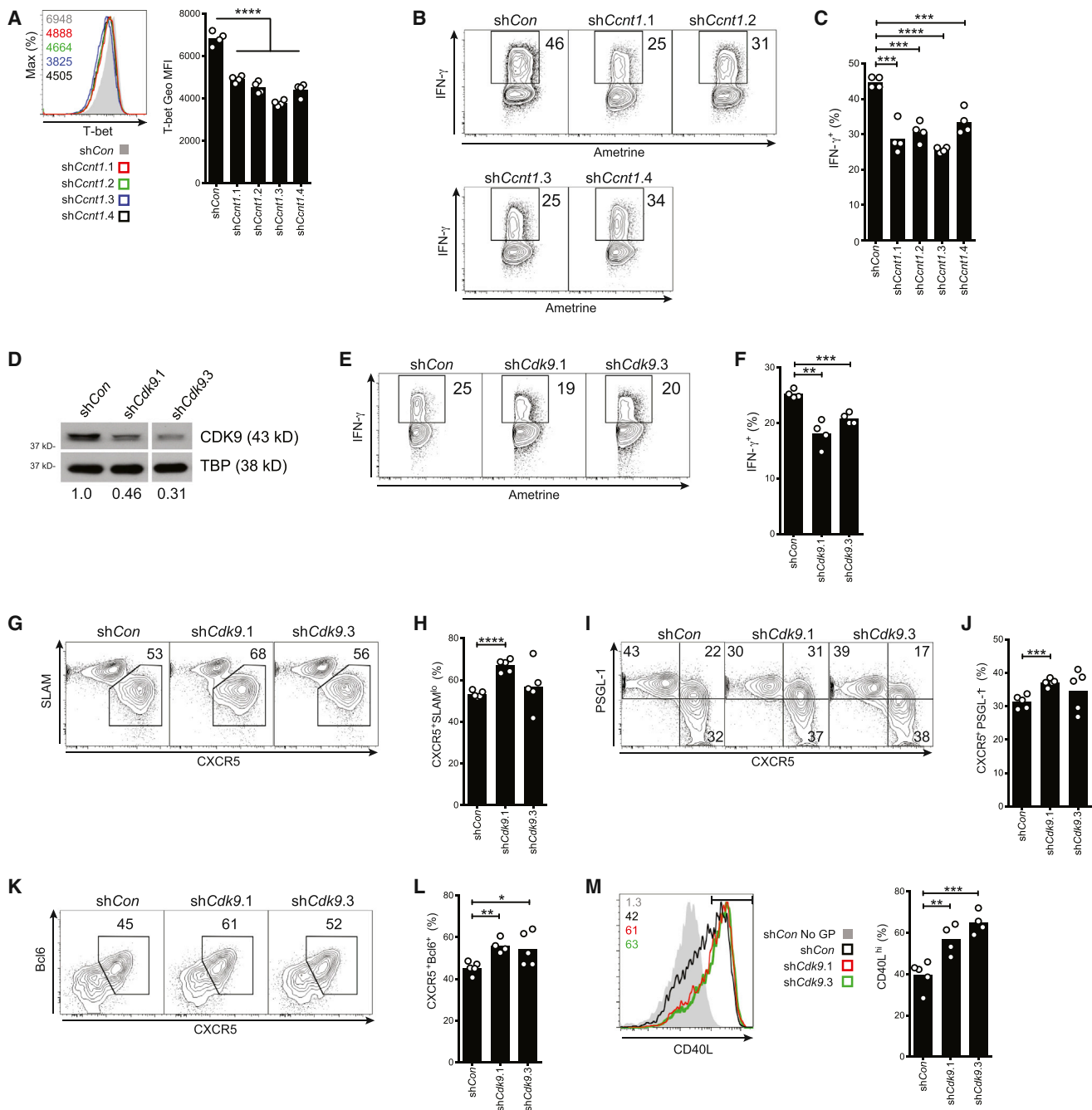


Figure 5. Cyclin T1 and Cdk9 Depletion Impairs Th1 Cell Differentiation In Vitro and In Vivo

(A–C, E, and F) CD4⁺ T cells were transduced with *Ccnt1* shRNAs and cultured under Th1-cell-biasing conditions for 4 days before restimulation with PMA and ionomycin for 1 (E and F) or 4 (A–C) hr.

(A) T-bet expression by shCcnt1⁺ CD4⁺ T cells. Quantitation and an example histogram are shown.

(B) Flow cytometry plots of IFN-γ expression by shCcnt1⁺ CD4⁺ T cells upon restimulation.

(C) Quantitation of (B), for all samples.

(D) Cdk9 protein expression in shCdk9⁺ MCC T cells.

(E) Flow cytometry plots of IFN-γ expression by shCdk9⁺ CD4⁺ T cells upon restimulation.

(F) Quantitation of (E), for all samples.

(G and H) Flow cytometry plots (G) and quantitation (H) of Tfh cell differentiation (CXCR5⁺SLAMF^{lo}) by shCdk9⁺ SMARTA CD4⁺ T cells at 6 days after LCMV infection.

(I and J) Flow cytometry plots (I) and quantitation (J) of GC Tfh cell differentiation (CXCR5⁺PSGL-1^{lo}) by shCdk9⁺ SMARTA CD4⁺ T cells at 6 days after LCMV infection.

(legend continued on next page)

approximately 1 week. Under these conditions, we estimated that each shRNA was represented ~500 times per mouse after T cell engraftment, which is ~10-fold higher representation of each shRNA than a recent pooled screen that examined skin cell development in vivo (Beronja et al., 2013; Zuber et al., 2011) and involved transferring 10-fold fewer T cells than a recent immunotherapy screen in T cells (Zhou et al., 2014). Taking these studies into account with our results, we anticipate that by using our system and applying deeper sequencing, it is likely to be feasible to perform phenotypic screens on pools of 1,000 or more shRNAs in parallel in T cells during infection.

The screen discovered unanticipated roles for Cyclin T1 and Cdk9 in the regulation of T cell differentiation during antiviral immune responses and emphasizes the specificity that ubiquitously expressed factors can have. Cyclin T1 and Cdk9 are two widely expressed components of P-TEFb (Oven et al., 2007; Peterlin and Price, 2006), which stimulates the transition of paused RNA polymerase II complexes into productive elongation (Peterlin and Price, 2006). The regulation of transcription elongation might govern a substantial fraction of differential expression of transcriptionally active genes (Min et al., 2011; Peterlin and Price, 2006; Rahl et al., 2010), and it is notable that this process is also critical in the regulation of HIV transcription in CD4⁺ T cells. The fact that depletion of Cyclin T1 or Cdk9 in activated T cells results in specific alterations in their differentiation in vitro and in vivo indicates that these factors are utilized in context-specific regulation of gene expression in T cells, despite their ubiquitous expression. Indeed, ChIP-seq analysis showed that Cyclin T1 is specifically recruited to subsets of genes, including *Tbx21*, *Prf1*, *Ifng*, and *Ii2ra*, that are activated in response to TCR-like stimulation of CD8⁺ T cells (M.E.P., unpublished data).

Given the known functions of Cyclin T1 and Cdk9 in P-TEFb, one interpretation of our results is that T cell differentiation is regulated via transcriptional elongation by P-TEFb. However, the phenotypes upon Cyclin T1 and Cdk9 depletion were not identical, although they were similar. One simple explanation of this outcome is that shRNA-mediated depletion of Cdk9 was less efficient than for Cyclin T1, resulting in different amounts of functional P-TEFb in each case. Another interpretation is alternative factors that “compensate” for reductions in wild-type Cyclin T1 or Cdk9 amounts, and which possess distinct activities or targeting, caused the observed phenotypes. Indeed, other Cyclins and Cyclin-dependent kinases can phosphorylate the C-terminal domain of RNA Pol II at serine 2 and regulate transcription, and could be cell type specific (Blazek et al., 2011). Finally, Cyclin T1 and Cdk9 could have roles in T cells apart from their established roles in P-TEFb. Future studies to elucidate the specific roles of Cyclin T1 and Cdk9 and how they integrate with the external signals that govern CD4⁺ and CD8⁺ T cell differentiation are likely to open previously unappreciated insights into T cell function. In summary, the functional genetic approach described here is likely to facilitate the identification of many previously unknown players.

EXPERIMENTAL PROCEDURES

Animals and Viruses

C57BL/6 (B6) mice were purchased from the Jackson Laboratory. CD45.1⁺ SMARTA (SM; lymphocytic choriomeningitis virus [LCMV] gp66-77-IA^b specific) (Oxenius et al., 1998) and Blimp1-YFP mice were bred in-house. LCMV gp33-41-specific P14 Thy1.1⁺ mice used for in vivo analysis were a gift from R. Ahmed (Emory University); P14 *Tcra*^{-/-} mice were used for in vitro studies (Taconic). All mice were maintained in specific-pathogen-free facilities and used according to protocols approved by the animal care and use committees of the LIAI and TSRI-FL. Virus stocks were made as described (Johnston et al., 2009), and LCMV titers in tissues were assessed by plaque assay.

Pooled Screening Approaches

Please refer to [Supplemental Experimental Procedures](#).

Flow Cytometry

Single-cell suspensions of spleens were prepared by mechanical disruption. Surface staining for flow cytometry was performed by standard techniques (Johnston et al., 2009) and the following clones: CD4 (RM4-5), CD45.1 (A20), CD44 (IM7), and CD62L (MEL-14) (eBiosciences); CD8 (53-6.7) and B220 (RA3-6B2) (BD Biosciences); as well as CD8 (53-6.7), CD127 (A7R34), KLRG-1 (2F1), CD90.1 (OX-7), and SLAM (TC15-12F12.2) (BioLegend). CXCR5 staining was performed as described (Choi et al., 2013). Intracellular staining after surface stains was performed using the “Foxp3 staining buffer” set (eBiosciences), using anti-Bcl6 monoclonal antibody (K112-91, BD Biosciences), anti-Tbet (4B10), or anti-Granzyme B (GB11) (Biolegend).

Adoptive Transfer Analysis of Individual shRNAmirs in CD8⁺ or CD4⁺ T Cells

For in vivo confirmation of “hits,” SMARTA CD4⁺ or P14 CD8⁺ T cells were transduced with viral supernatants generated from individual shRNAmir-RV constructs ([Supplemental Experimental Procedures](#)). A total of 5 × 10⁵ P14 cells were transferred into 6-week-old B6 mice 1 or 2 days after activation and analyzed on day 7 or 8 after infection. Note, transfer of P14 cells on day 1 rather than day 2 after activation was found to recapitulate differentiation more physiologically (data not shown). For CD4⁺ T cell experiments, 25,000 SMARTA cells were transferred.

RNA and Protein Analysis

Total RNA was isolated from transduced (Ametrine⁺) CD4⁺ or CD8⁺ T cells and used for cDNA synthesis as previously described (Johnston et al., 2009). qPCR reactions were performed in triplicate using the SYBR Select Master Mix (Life Technologies) on a Roche Lightcycler 480, using primers specific to *Prdm1* (F-5'-TTCTCTTGGAAAACGTGTGGG-3'; R-5'-GGAGCCGGAGCTAGACTTG-3') and *Tbx21* (F-5'-ACCAACAACAAGGGGGCTTC-3'; R-5'-CTCTGGCTCTCCATCATTACC-3'). For immunoblot analysis, whole-cell lysates were obtained from CD8⁺ T cells on day 6 after activation, from CD4⁺ T cells 5 days after activation, or from MCC-T cells by sorting transduced (Ametrine⁺) cells and lysis in 150 mM NaCl, 25 mM Tris (pH 7.5), 1% Triton X-100, 0.1% SDS, 0.5% Deoxycholate, and complete protease inhibitors (Roche). 25 μg of protein was resolved by 8% SDS-PAGE, transferred to nitrocellulose membranes, and probed with anti-Cyclin T1 (sc-10750), anti-Blimp1 (sc-47732), anti-Cdk9 (sc-484) (Santa Cruz Biotechnology), anti-Perforin (ab16074), and anti-beta Actin (Abcam ab8227).

SUPPLEMENTAL INFORMATION

Supplemental Information includes seven figures, one table, and Supplemental Experimental Procedures and can be found with this article online at <http://dx.doi.org/10.1016/j.immuni.2014.08.002>.

(K and L) Flow cytometry plots (K) and quantitation (L) of CXCR5 and Bcl6 expression by shCdk9⁺ SMARTA CD4⁺ T cells, 4 days after LCMV infection.

(M) Histograms of CD40L expression on shCdk9⁺ SMARTA CD4⁺ T cells, after isolation from spleens 4 days after LCMV infection and restimulation with GP₆₁₋₈₀ peptide for 4 hr. The percentages of CD40L^{hi} SMARTA are shown and summarized (right).

Data are representative of two independent experiments. *p < 0.05, **p < 0.01, ***p < 0.001, ****p < 0.0001.

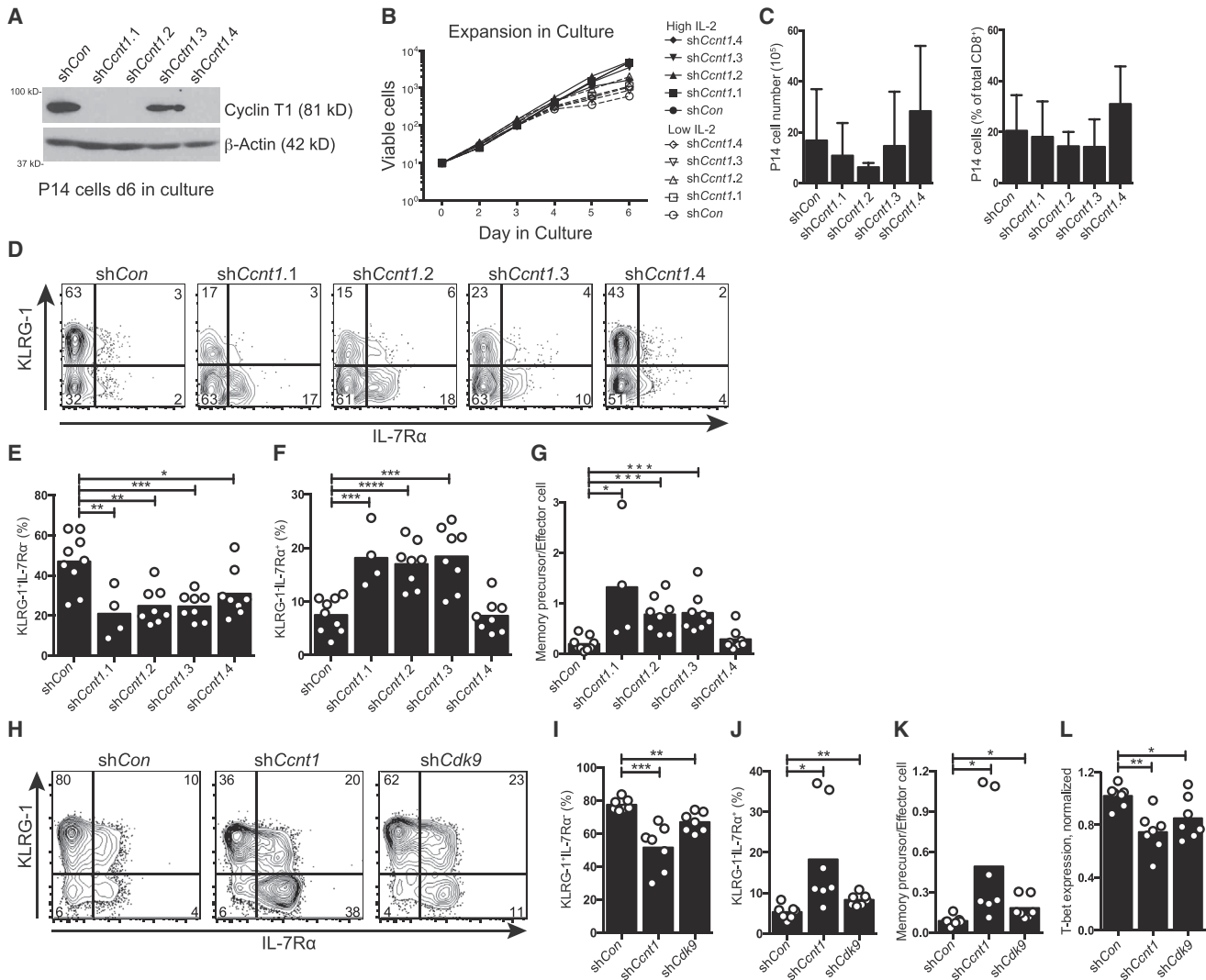


Figure 6. Cyclin T1 and Cdk9 Depletion Impairs Generation of Effector CD8⁺ T Cells during LCMV Infection

(A) Immunoblot analysis of Cyclin T1 in FACS-sorted shCcnt1⁺ P14 CD8⁺ T cells. Cells were cultured 6 days in low IL-2 (10 U/ml).

(B) Expansion of FACS-sorted shCcnt1⁺ P14 CD8⁺ T cells in culture. Low IL-2 (10 U/ml), high IL-2 (100 U/ml).

(C–L) Adoptively transferred P14 CD8⁺ T cells transduced with the indicated shRNAs were analyzed on day 7 (C–G) or day 8 (H–L) after LCMV infection.

(C) The numbers and percentages of shCcnt1⁺ P14 cells in the spleen.

(D) Contour plots show KLRG-1 and IL-7R α staining on shCcnt1⁺ P14 CD8⁺ T cells from representative mice 7 days after LCMV infection.

(E–G) Quantitation of CD8⁺ T cell subsets from shCcnt1⁺ P14 cells in vivo.

(E) Short-lived effector cells (KLRG-1^{hi}IL-7R α ^{lo}).

(F) Memory precursor cells (KLRG-1^{lo}IL-7R α ^{hi}).

(G) Ratio of memory precursor to short-lived effector phenotype P14 cells, per mouse.

(H) Contour plots show KLRG-1 and IL-7R α staining by shCdk9⁺ P14 CD8⁺ T cells from representative mice 8 days after LCMV infection.

(I–L) Quantitation of CD8⁺ T cell subsets from shCdk9⁺ P14 cells in vivo.

(I) Short-lived effector cells (KLRG-1^{hi}IL-7R α ^{lo}).

(J) Memory precursor cells (KLRG-1^{lo}IL-7R α ^{hi}).

(K) Ratio of memory precursor to short-lived effector phenotype P14 cells, per mouse.

(L) Summarized T-bet expression based on intracellular staining and flow cytometry.

Each symbol represents T cells from separate mice. Data are pooled from two independent experiments. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$. Error bars represent standard deviations.

AUTHOR CONTRIBUTIONS

R.C. established the pooled approaches, implemented and analyzed the CD8⁺ T cell screen and follow-up, and assisted writing the paper. S.B. designed, implemented, and analyzed the CD4⁺ T cell screen and follow-up,

and assisted writing the paper. M.A.F. performed CD8⁺ T cell follow-up analysis. B.L. designed the sequencing bioinformatic pipeline. R.J.J. designed vectors and did preliminary experiments. S.S., N.X., and Y.-C.L. provided reagents and advice. A.R. assisted with experimental design, provided resources, and assisted writing the paper. B.P. designed the sequencing

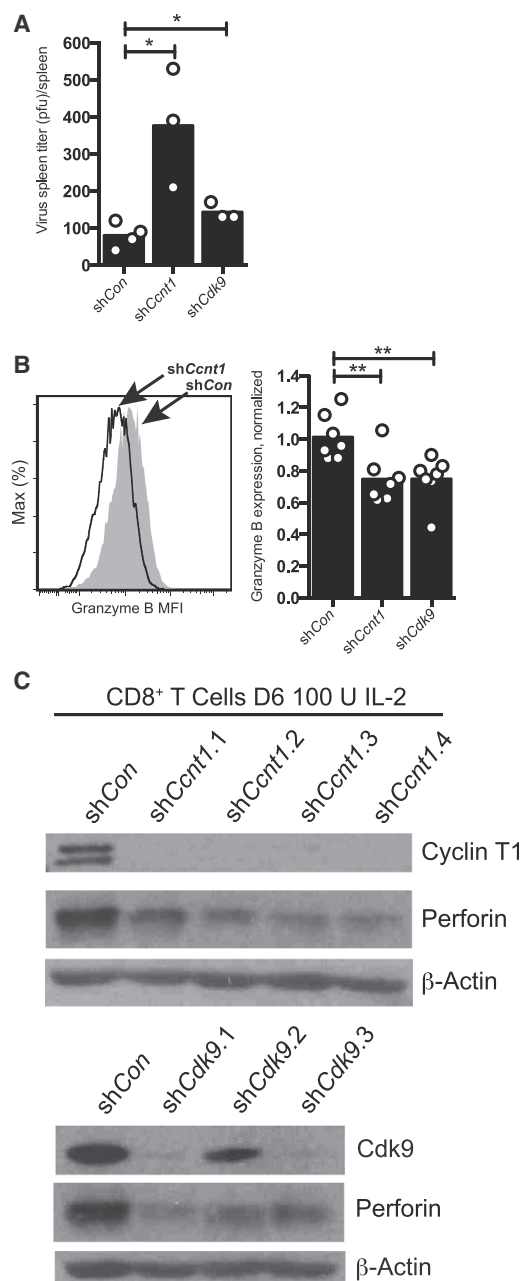


Figure 7. Cyclin T1 and Cdk9 Are Required for Antiviral CTL Functions

(A) LCMV titers in spleen were determined 8 days after LCMV infection. (B) Granzyme B expression in P14 cells at day 8 postinfection. Histogram (left) and quantitation (geometric MFI; right). (C) Immunoblot analysis of Cyclin T1, Cdk9, Perforin, and β -actin expression in whole-cell lysates from flow cytometry-sorted shCcnt1⁺ and shCdk9⁺ P14 CD8⁺ T cells after 6 days in culture (100 U/ml).

analysis pipeline and supervised statistical analyses. M.E.P. and S.C. conceived of the studies, designed experiments, analyzed data, supervised the projects, and wrote the paper.

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Supplemental Information

In Vivo RNA Interference Screens

Identify Regulators of Antiviral

CD4⁺ and CD8⁺ T Cell Differentiation

Runqiang Chen, Simon Bélanger, Megan A. Frederick, Bin Li, Robert J. Johnston, Nengming Xiao, Yun-Cai Liu, Sonia Sharma, Bjoern Peters, Anjana Rao, Shane Crotty, and Matthew E. Pipkin

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In vivo RNA interference screens identify regulators of antiviral CD4 and CD8 T cell differentiation

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Supplemental Figure S1. Modification of an shRNAmir-RV prevents rejection of transduced T cells after adoptive transfer and mediates efficient knockdown *in vivo* (related to Figure 1).

Supplemental Figure S2. Optimizations of retroviral transduction of CD8 T cells and of next generation sequencing libraries (related to Figure 1).

Supplemental Figure S3. The number of transferred P14 cells as well as the strain and dose of LCMV was optimized for the CD8 T cell screen (related to Figure 1).

Supplemental Figure S4. Activated and transduced P14 cells undergo effector CD8 T cell differentiation that is similar to naïve P14 cells following adoptive transfer and LCMV infection (related to Figure 1)

Supplemental Figure S5. Cloning strategies to mobilize shRNAmir pools into LMPd (related to Figure 2).

Supplemental Figure S6. Ccnt1-specific shRNAs inhibit expression of Cyclin T1 and IFN γ and promote Tfh formation (related to Figure 4).

Supplemental Figure S7. The P-TEFb subunit Cdk9 regulates Tfh formation (related to Figure 5).

Supplemental Table S1. Table of individual shRNAmirs in the TF-30 library (related to Figure 2).

Supplemental Experimental Procedures

Supplemental CD8 Screening Protocol

Supplemental CD4 Screening Protocol

Supplemental CD8 Sequencing Protocol

Supplemental CD4 Sequencing Protocol

Supplemental References

Figure S1

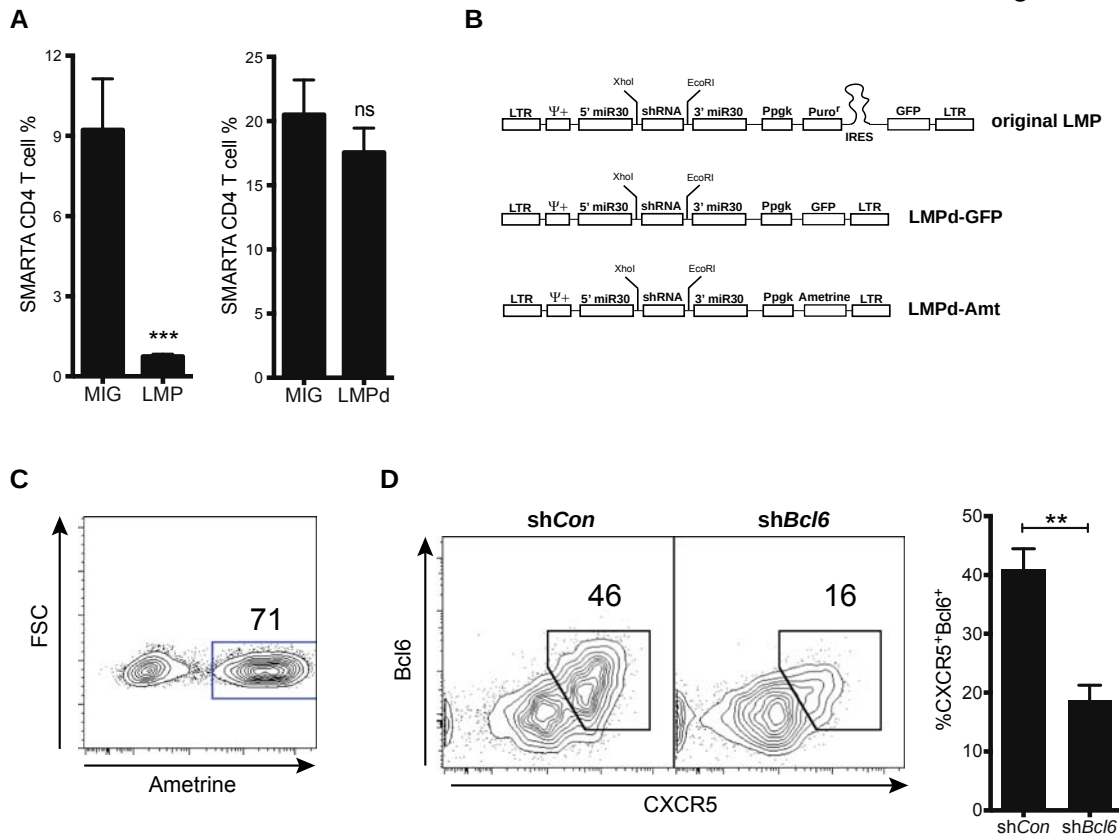


Figure S1: Modification of an shRNAmir-RV prevents rejection of transduced T cells after adoptive transfer and mediates efficient knockdown *in vivo* (related to Figure 1). **(A and B)** SMARTA CD4 T cells were transduced with MIG-RV, LMP-RV or LMPd-RV, adoptively transferred into B6 mice, and analyzed 8 days after LCMV infection. **(A)** Quantitation shows SMARTA CD4 T cells as a percentage of total CD4 T cells. **(B)** Schematic representations of the LMP and LMPd vectors. **(C and D)** SMARTA CD4 T cells were transduced with LMPd-RV encoding *Con* or *Bcl6* shRNAmir, transferred into B6 mice and analyzed 6 days after LCMV infection. **(C)** Representative FACS plot of Ametrine (Ai et al., 2008) expression in transduced CD4 T cells. **(D)** Representative flow cytometry plots of CXCR5 and Bcl6 expression by SMARTA CD4 T cells transduced with shCon or shBcl6 shRNAmir-RV (left panel). CXCR5⁺Bcl6⁺ cells among total SMARTA CD4 T cells were quantified (right panel). *** P<0.001. Error bars indicate standard deviations.

Figure S2

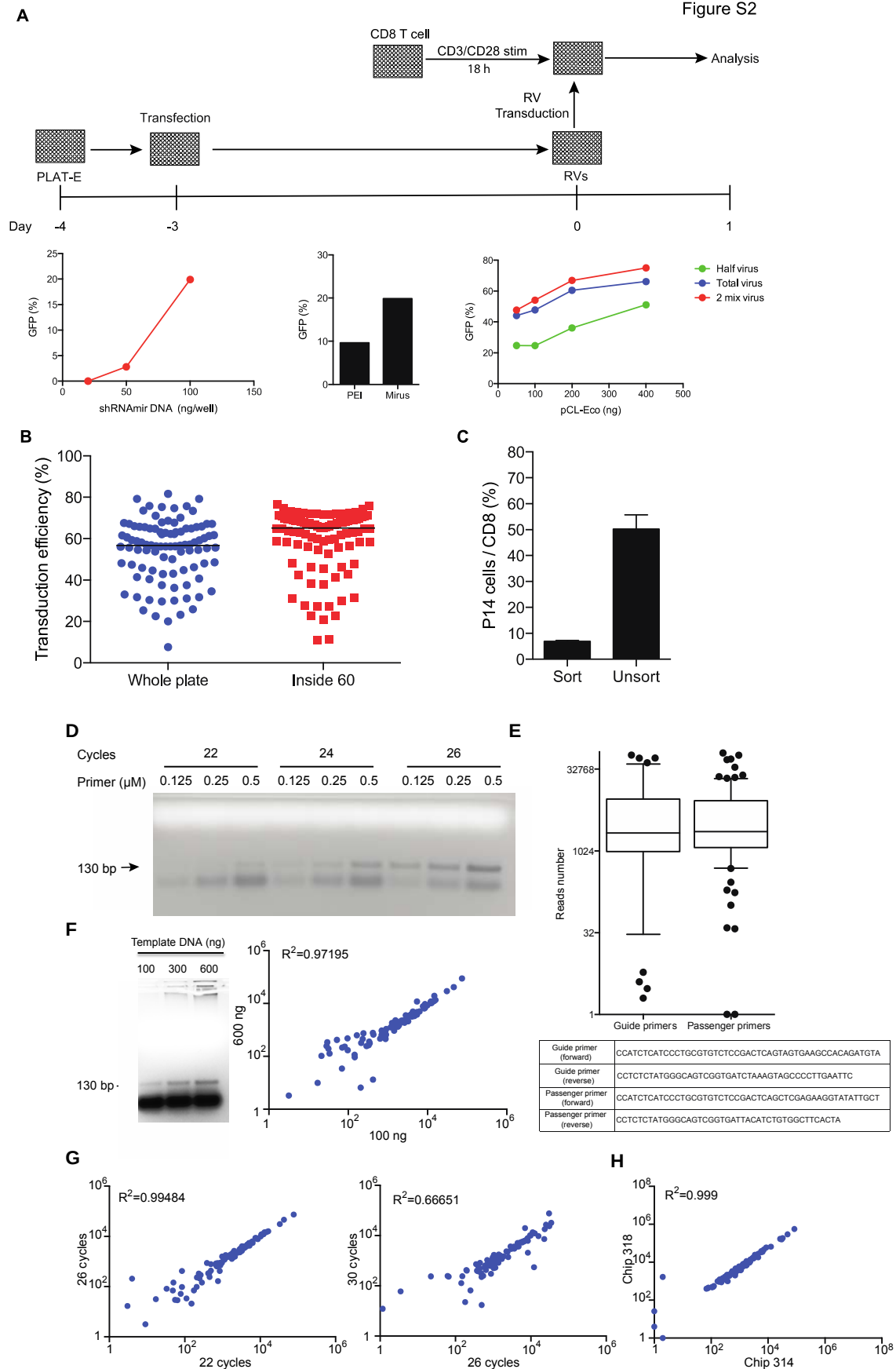


Figure S2. Optimizations of retroviral transduction of CD8 T cells and of next generation sequencing libraries (related to Figure 1). **(A)** P14 CD8 T cells were activated in 96-well plates pre-coated with anti-CD3 and anti-CD28 antibodies. One day later, cells were transduced with empty shRNA_{mir}-RV. Transduction efficiency was determined by assessing the frequency of GFP⁺ cells by flow cytometry one day after transduction. P14 CD8 T cells were transduced with RV prepared by transfection of PLAT-E with increasing amounts of shRNA_{mir} DNA (left), with different transfection reagents (middle) or with increasing amounts of the packaging vector pCL-Eco (right). In addition, P14 CD8 T cells were transduced with different volumes of RV supernatant (right). A representative experiment is shown out of two. **(B)** P14 CD8 T cells were plated in either a whole plate or in the inside 60 wells of 96-well plates coated with anti-CD3 and anti-CD28 antibodies followed with transduction with shRNA_{mir} RV. One day after the transduction, the frequency of A_{me}⁺ cells was determined by flow cytometry. **(C)** P14 CD8 T cells were activated in vitro with anti-CD3 and anti-CD28 followed by transduction. Transduced cells were either sorted into GFP⁺ cells or left unsorted prior to transfer in B6 mice. The total number of splenic P14 and endogenous CD8 T cells were analyzed 5 days after infection with 1.5×10^5 PFU LCMV clone 13. **(D)** Sequencing libraries (130 bp) were generated by PCR amplification of proviral sequences in genomic DNA from pooled transduced cells using different primer concentrations and numbers of PCR cycles. **(E)** Sequencing reads number distribution in shRNA_{mir} libraries from “input” generated with primers specific for the guide strand or for the passenger strand. **(F)** shRNA_{mir} libraries (130 bp) were generated by PCR amplification from the indicated amounts of template DNA and were sequenced independently. The scatter plot of sequencing reads for each shRNA_{mir} in libraries prepared with 100 ng or 600 ng of template DNA is shown. **(G)** The effects of extensive PCR is shown in scatter plots of sequencing reads for each shRNA_{mir} in libraries prepared with 22, 26 or 30 cycles of PCR amplification. **(H)** Scatter plot of sequencing reads for each shRNA_{mir} in a library sequenced with the Ion 314 Chip or with the Ion 318 Chip. The correlation coefficient, R^2 , is indicated.

A

CD8 T cell
CD3/CD28 stim
Day -2
0
4
Analysis

LCMV Armstrong
CD8 T cells
Analysis

Total P14 cells (10^7)

Endogenous CD8 T cells (10^7)

Transferred P14 cells

0 2x10⁵ 5x10⁵ 1x10⁶

0 2x10⁵ 5x10⁵ 1x10⁶

B

CD8 T cell
CD3/CD28 stim
Day -2
0
4
Analysis

LCMV Armstrong
CD8 T cells
Analysis

P14 cells (% of total blood CD8)

LCMV (10⁵ PFU)

Transferred P14 cells (10⁵)

0 0 0 0 2 5 20 100

(Naive P14) In vitro activated

0 5 5 5 5 5 5

C

CD8 T cell
CD3/CD28 stim
Day -2
0
2 3 4 5 6
Analysis

LCMV Arm or Clone13
5x10⁵ CD8 T cells
Analysis

P14 cells (10^5)

Days Post Infection

• Armstrong
• Clone 13

2 3 4 5 6

D

Day 4 Day 5 Day 6 Day 7

Uninfected

CD45.2

CD8

3.67 5.72 5.1 5.67

1.01 29 34.2 21.1

LCMV

Figure S3. The number of transferred P14 cells as well as the strain and dose of LCMV was optimized for the CD8 T cell screen (related to Figure 1). **(A)** P14 CD8 T cells were activated in vitro with anti-CD3 and anti-CD28 for 2 days. Following activation, cells were transferred into B6.SJL mice, and immediately infected with 2×10^5 PFU LCMV Armstrong 5. The total number of splenic P14 and endogenous CD8 T cells were analyzed 4 days after infection. Data shown as mean of two mice per group. **(B)** P14⁺ Thy1.1⁺ CD8 T cells were activated in vitro and transferred into B6 (Thy1.2⁺) mice followed by infection with increasing doses of LCMV Armstrong 5. The number of P14 CD8 T cells in the blood was determined 4 days after infection. Data shown as mean $\pm$ SD of three mice per group. The percentage of P14 cells in the blood increased substantially in a fashion dependent on the increased dose of LCMV-Armstrong indicating that a stronger LCMV infection resulted in a larger P14 cell response. **(C)** P14⁺ Thy1.1⁺ CD8 T cells were activated in vitro and transferred into B6 (Thy1.2⁺) mice. Mice were infected with LCMV Armstrong 5 (1×10^7 PFU) or LCMV clone 13 (1.5×10^5 PFU). Splenic P14 CD8 T cell number was determined at different time points after infection.

Data shown as mean from three mice per group. P14 cells from LCMV clone 13 infected mice accumulated most strongly, suggesting that this infection was more robust than infection with high-dose LCMV Armstrong, and was less efficiently curtailed by the transfer of activated P14 cells (**D**) The frequency of CD45.2⁺ P14 CD8 T cells was determined in the spleens of B6.SJL mice at the indicated time points following infection with 1.5×10^5 PFU LCMV clone13 or no infection. Representative dot plot graphs are shown. Error bars indicate standard deviations.

Figure S4

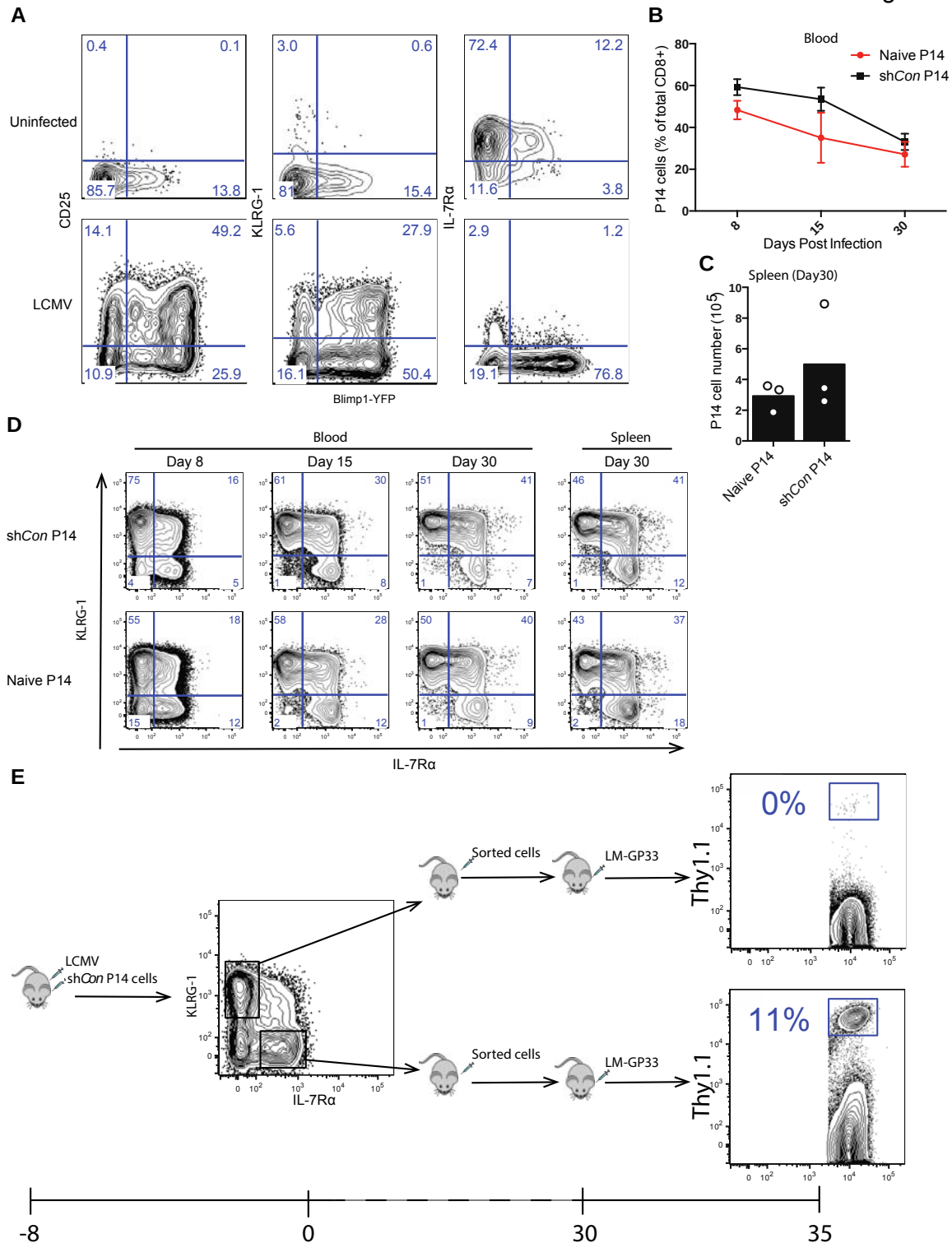


Figure S4. Activated and transduced P14 cells undergo effector CD8 T cell differentiation that is similar to naïve P14 cells following adoptive transfer and LCMV infection (related to Figure 1). **(A)** Flow cytometry shows staining of the indicated markers relative to Blimp1-YFP reporter expression of *in vitro*-activated and adoptively transferred P14 cells from representative infected and uninfected mice 5 days after transfer and infection. **(B)** Naïve P14 CD8 T cells (5,000) or P14 CD8 T cells transduced with shCon (500,000) were adoptively transferred into B6 mice, and immediately

infected with 1.5×10^5 PFU LCMV clone13. The number of P14 CD8 T cells in the blood was determined at different time points after infection. Similar percentages of P14 cells (~50-60%) were found in the blood and in the spleen on day 8 following infection in both groups. (C) The total number of splenic P14 CD8 T cells was analyzed 30 days after infection. Data shown as mean of three mice per group. (D) Representative contour plots of KLRG-1 and IL-7R α expression in Naïve P14 CD8 T cells or P14 CD8 T cells transduced with shCon at different time points after LCMV infection. P14 cell numbers from each group contracted over time, but were maintained comparably between both groups in these tissues until days 15 and 30. Both naïve, as well as activated and transduced, P14 cells resulted in selective accumulation of IL-7R α^{hi} cells during transition into the memory phase; both groups showed a progressive decrease in the percentage of KLRG-1 $^{\text{hi}}$ IL-7R α^{lo} cells, and a concomitant increase in both KLRG-1 $^{\text{hi}}$ IL-7R α^{hi} and KLRG-1 $^{\text{lo}}$ IL-7R α^{hi} cells during this time. (E) To confirm the functional potentials of these different subsets, KLRG-1 $^{\text{hi}}$ IL-7R α^{lo} and KLRG-1 $^{\text{lo}}$ IL-7R α^{hi} cells derived from P14 cells that had been activated and transduced with shCon-RV prior to adoptive transfer and infection were sorted from mice on day 8 post infection, and transferred to naïve B6 recipients. P14 CD8 T cells were transduced with shCon, transferred in 6-week-old B6 mice, and immediately infected with 1.5×10^5 PFU LCMV clone13. 8 days after infection, P14 CD8 T cells were isolated by positive selection using CD90.1 magnetic beads, and CD90.1 $^+$ cells were sorted into effector cells (KLRG-1 $^{\text{hi}}$ IL-7R α^{lo}) and memory precursor cells (IL-7R α^{hi} KLRG-1 $^{\text{lo}}$). Equal numbers (7×10^5) of effector cells or memory precursor cells were adoptively transferred into B6 mice. 30 days after transfer, mice were infected with 1×10^5 CFU of LM-GP33. The percentage of P14 cells in total CD8 T cells were analyzed 5 days after infection. P14 cells derived from transfer of KLRG-1 $^{\text{lo}}$ IL-7R α^{hi} expanded strongly by day 5 after LM-GP33 challenge, whereas P14 cells derived from transfer of KLRG-1 $^{\text{hi}}$ IL-7R α^{lo} cells expanded weakly, confirming the appropriate functionality of the cells. Error bars indicate standard deviations.

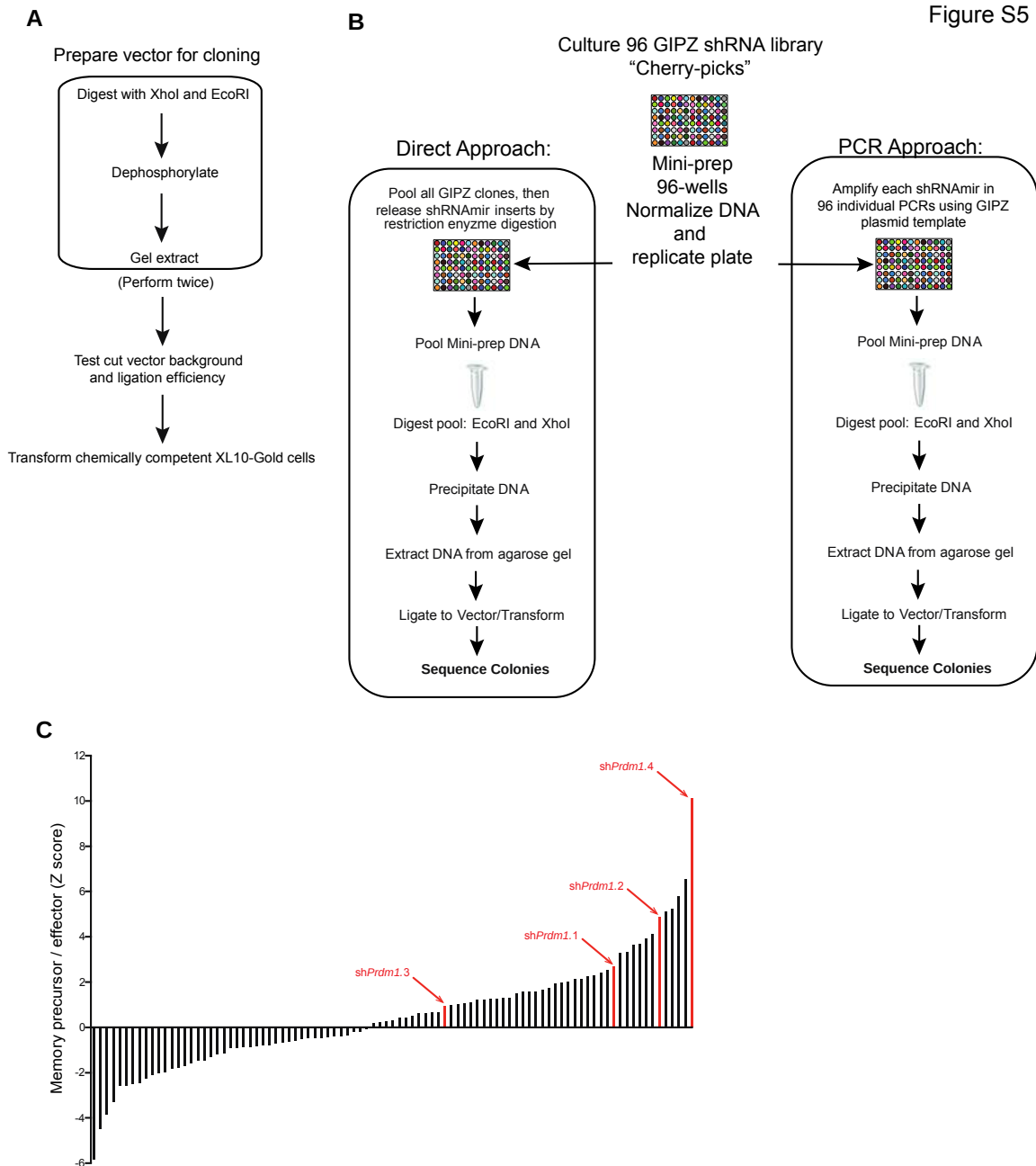


Figure S5. Cloning strategies to mobilize shRNAmir pools into LMPd (related to Figure 2). **(A)** LMPd-Ametrine vector was digested with EcoRI and XhoI twice. After purification, the cut vector background and ligation efficiency were tested. **(B)** 96 shRNAmirs DNA from GIPZ library were subcloned using two approaches: Direct and PCR Approaches. In the Direct approach, 96 shRNAmirs were pooled, digested with EcoRI and XhoI and purified. In the PCR approach, 96 shRNAmirs were individually amplified with PCR, then PCR products were pooled, digested with EcoRI and XhoI and purified. The purified shRNAmir fragments were ligated into EcoRI/XhoI-digested LMPd-Ametrine vector and transformed. 768 colonies were picked and sequenced. **(C)** Z-score values of all shRNAmirs in the first RNAi screen in CD8 T cells. shRNAmirs specific for *Prdm1* are highlighted in red.

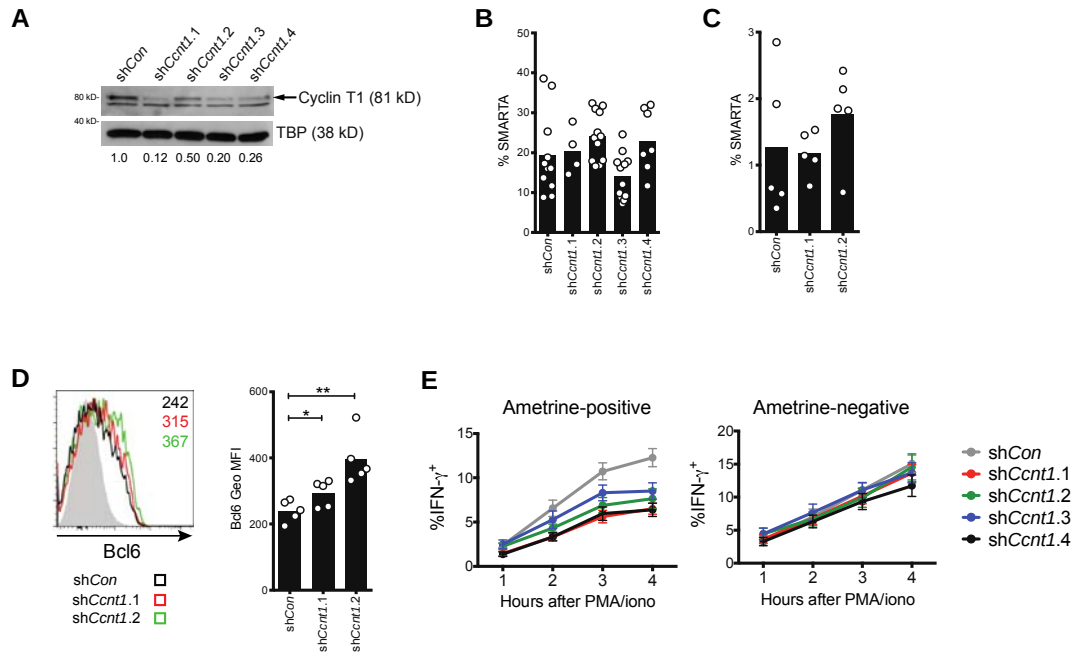


Figure S6. Ccnt1-specific shRNAs inhibit expression of Cyclin T1 and IFN γ and promote Tfh formation (related to Figure 4). **(A)** Expression of Cyclin T1 protein in WT CD4 T cells transduced with the indicated shRNAmir. Numbers below each lane indicate the ratio of Cyclin T1 to TBP relative to that of the control shRNAmir. **(B)** Quantitation of shRNAmir-transduced SMARTA CD4 T cells as a percentage of CD4 T cells 6 days after LCMV infection. **(C)** Quantitation of shRNAmir-transduced SMARTA CD4 T cells as a percentage of CD4 T cells 3 days after LCMV infection. **(D)** Bcl6 expression and geometric MFI quantitation in SMARTA transduced with the indicated shRNAmirs 4 days after LCMV infection. The numbers in the overlaid histograms indicate the geometric MFI of Bcl6 in SMARTA. **(E)** shRNAmir-transduced WT CD4 T cells were cultured under neutral conditions for 4 days before restimulation with PMA/ionomycin for the indicated times before staining for IFN- γ . Quantification of the IFN- γ ⁺ CD4 in the Ametrine-negative and -positive populations at the indicated times is shown. Each symbol represents T cells from an individual mouse. Data is representative of 2 independent experiments. *P<0.05, **P<0.01. Error bars indicate standard deviations.

Figure S7

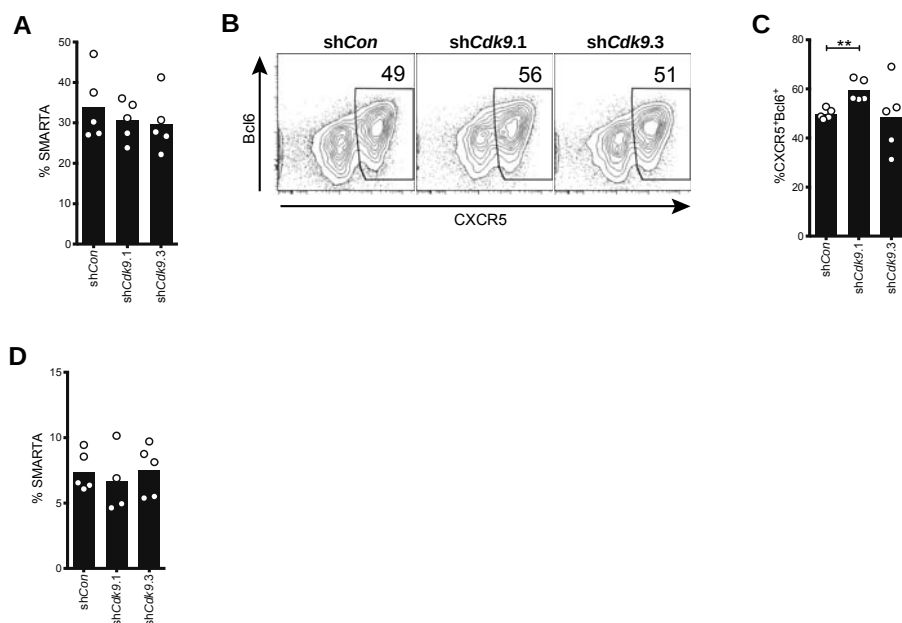


Figure S7. The P-TEFb subunit Cdk9 regulates Tfh formation (related to Figure 5). **(A)** Quantitation of shRNAmir-transduced SMARTA CD4 T cells as a percentage of CD4 T cells 6 days after LCMV infection. **(B)** Representative FACS plots of CXCR5 and Bcl6 expression in SMARTA CD4 T cells transduced with the indicated shRNAmir 6 days after LCMV infection. **(C)** Percentages of CXCR5⁺Bcl6⁺ Tfh in shRNA-transduced SMARTA cells. **(D)** Quantitation of shRNAmir-transduced SMARTA CD4 T cells as a percentage of CD4 T cells 4 days after LCMV infection. Data is representative of 2 independent experiments. **P<0.01, ***P<0.001.

Table S1. Table of individual shRNAmirs in the TF-30 library (related to Figure 2).

[illegible]

Supplemental Experimental Procedures

Large-scale sub-cloning of shRNAmirs into retroviral vectors. To generate the “TF-30” mini-library we subcloned shRNAmir inserts from 96 clones derived from commercially available libraries (Thermo Fisher Scientific, Inc. and TransOmic Technologies, Inc.) into our pLMPd-Amt vector (see Results) using pooled strategies. The majority of clones were selected initially from the GIPZ mouse shRNA whole-genome library (Thermo Fisher). The remaining clones were top-scoring shRNAmir designs from shERWOOD analysis (TransOmic Technologies, Inc.), an algorithm trained on data from a high-throughput functional shRNA-sensor assay (Fellmann et al., 2011). These clones are highlighted in blue (**Table S1**). More reliable RNAi was achieved using shERWOOD designed shRNAmirs compared to other commercially available shRNAmirs for a randomly selected subset of genes (M.E.P. and S.C., unpublished data).

We compared two approaches for subcloning all the shRNAmir inserts from a single pool (**Figure S5A and S5B**). In the first approach, purified plasmid DNA from the original library clones was pooled and cleaved with XhoI and EcoRI to release all 96 inserts simultaneously. In the second approach, each individual shRNAmir insert was amplified in 96 parallel PCR reactions from purified plasmid GIPZ clone DNA using primers specific for common regions in the flanking miR-30 sequences, and that included restriction enzyme sequences (XhoI and EcoRI) compatible with directional cloning (restriction enzymes sequences are underlined: XhoI-5'mir30, CAGAAGGCTCGAGAAGGTATATTGCTGTTGACAGTGAGCG; EcoRI-3'mir30, CTAAAGTAGCCCCTTGAATTCCGAGGCAGTAGGCA). All shRNAmir were amplified in 20 μ l reactions containing 5 ng of plasmid DNA, 0.4 U of Phusion Hot Start II High-Fidelity DNA polymerase (Thermo Scientific), 1x Phusion HF buffer, 0.5 μ M of each primer and 0.2 mM of each dNTPs using the following cycling parameters: 94 °C for 5 min; 16 cycles of 94 °C for 30 s, 54 °C for 30 s and 75 °C for 30 s; 75 °C for 2 min.; amplification of each shRNAmir was confirmed by agarose gel electrophoresis, and the PCR products were pooled and digested with DpnI to destroy remaining template plasmid DNA, and then digested again with EcoRI and XhoI to generate sticky ends (**Figure S6**). In both approaches, the inserts were gel purified after restriction enzyme digestion using Qiaquick columns. Inserts prepared with each method were ligated separately into 100 ng of pLMPd-Amt in a 2.5:1 molar ratio (Rapid Ligation Kit, Thermo Scientific), and 10% of each ligation was transformed into 50 μ l of XL10-Gold chemically competent cells (Agilent Technologies). 384 colonies from each

transformation were picked and cultured in 96-well plates to prepare glycerol stocks, which were sent for standard DNA sequencing (Beckman Genomics, Inc.). The BLAST tool was used align all reads to a reference file built from the 96 shRNA sequences we had selected to clone from the original GIPZ library. Each cloning approach resulted in recovery of ~85% of intended shRNAmirs with similar fidelity after sequencing 384 clones, and 100% of shRNAmirs were recovered as perfect matches when clones from both cloning approaches were considered together (data not shown). All constructs are freely available upon request.

CD8⁺ T cell isolation and stimulation. P14 CD8⁺ T cells were isolated from whole splenocytes by negative selection using magnetic beads (Invitrogen) and resuspended in D-10 (DMEM + 10% FCS (fetal calf serum), supplemented with 2 mM GlutaMAX (Gibco), 100 U/mL Penicillin/Streptomycin (Gibco), and 50 μ M BME). 2×10^5 naïve CD8⁺ T cells were stimulated with anti-CD3 (clone 2C11) and anti-CD28 (clone 37.51) (1 μ g/ml) in 96-well plates pre-coated with 100 μ g/mL goat anti-hamster IgG (H+L) secondary antibody (Pierce).

CD8 T cell transduction and adoptive transfer. shRNAmir-RVs were packaged in a 96-well plate format. MicroRNA-adapted short hairpin RNA expressing retroviral vectors (pLMPd-Amt) were used to generate retrovirus from the PLAT-E cell line. PLAT-E cells were seeded in a 96-well plate (6×10^4 cells/well) 1 day before the transfection. The following morning, each well was transfected with 0.2 μ g of shRNAmir vector and 0.2 μ g of pCL-Eco using TransIT-LT1 (Mirus). Each well of PLAT-E was transfected with a different shRNAmiR construct. Thus, for the shRNAmir library described in this manuscript, the 96 constructs were transfected individually in 96 wells of PLAT-E (**Figure 1B**). 36 and 48 hours after the transfection, 70 μ l of the RV-containing supernatant was harvested, stored at 4°C and replaced with fresh medium. 60 hours after transfection, supernatant was harvested for a third time, combined with the stored supernatant and used for transductions. RV supernatant from an individual well was used to transduce CD8 T cells. 96 independent RV supernatants were done in parallel (**Figure 1B**). This was done to ensure that CD8 T cells were transduced with a single shRNAmir construct. Plates containing stimulated CD8 T cells were spun down, and the medium was removed and replaced with RV supernatant supplemented with 50 μ M BME and 8 μ g/mL polybrene (Millipore) followed by spinfection (60 min. centrifugation at 2000 rpm and 37°C). After transduction, the RV-containing medium was replaced with D-10. 24 hours after the transduction, cells were pooled and 5×10^5 P14 cells were transferred into B6 mice (**Figure 1B**). The remaining cells were sorted based

on the top 20% Ametrine1.1 MFI of Amt⁺ cells as the “Input” sample (**Figure 1B**), DNA was isolated using the FlexiGene DNA kit (Qiagen). For analysis of target gene knockdown, naïve P14 cells were activated in vitro and infected with concentrated shRNAmir-RV supernatants as described (Pipkin et al., 2010). 24 hours after infection, transduced cells were sorted to >95% purity by FACS and cultured until day 6 in media containing either 10 or 100 U/mL rhIL-2 exactly as described (Pipkin et al., 2010).

In vivo CD8 T cell procedures. Mice were infected with 1.5×10^5 plaque forming units (PFU) of LCMV clone 13, within 1 hour after cell transfer. 7 days after infection, spleens from 8 recipient mice were combined and CD8 T cells from whole splenocytes were isolated by positive selection using CD90.1 magnetic beads (Miltenyi) and donor P14 CD8 were sorted based on the top 20% Ametrine1.1 MFI of Amt⁺ cells using a BS FACS Aria (BD Biosciences), then based on KLRG1 and CD127 expression, top 20% of Amet⁺ cells were sorted into effector and memory precursor populations with effector cells defined as KLRG1^{hi}CD127^{lo} and memory precursor as CD127^{hi}KLRG1^{lo} cells (**Figure 1B**). 8×10^5 KLRG1^{hi}CD127^{lo} effector cells and 1.1×10^5 CD127^{hi}KLRG1^{lo} memory precursor cells were routinely obtained after sort. DNA was isolated from both populations using the FlexiGene kit (Qiagen).

CD4 T cell isolation and stimulation. SMARTA CD45.1⁺ CD4 T cells were isolated from whole splenocytes by negative selection using magnetic beads (Miltenyi) and resuspended in D-10 (DMEM + 10% FCS (fetal calf serum), supplemented with 2 mM GlutaMAX (Gibco) and 100 U/mL Penicillin/Streptomycin (Gibco)) with 50 μ M BME and 2 ng/ml mIL-7 (PeproTech). 2×10^5 naïve CD4 T cells per well were stimulated in 96-well plates pre-coated with 8 μ g/mL anti-CD3 (clone 2C11, BioXcell) and anti-CD28 (clone 37.51, BioXcell). Cells were plated in the center of two 96-well plates (48 wells in each plate) to avoid the edges of the plates. Empty wells were filled with D-10 medium.

CD4 T cell transduction and adoptive transfer. First, shRNAmir-RVs were packaged in a 96-well plate format. MicroRNA-adapted short hairpin RNA expressing retroviral vectors (pLMD-Amt) were used to generate retrovirus from the PLAT-E cell line. PLAT-E cells were seeded in a 96-well plate (4×10^4 cells/well) 1 day before the transfection. The following morning, each well of PLAT-E was transfected with 0.2 μ g of shRNAmir vector and 0.1 μ g of pCL-Eco using polyethylenimine (Sigma-Aldrich). Each well of PLAT-E was transfected with a different shRNAmir construct. Thus, for the shRNAmir library described in this manuscript, the 96 constructs were transfected individually in 96

wells of PLAT-E (**Figure 3A**). 24 hours after the transfection, the RV-containing supernatant was harvested, stored at 4°C and replaced with fresh medium. 48 hours after transfection, supernatant was harvested for a second time, combined with the stored supernatant and used for transductions.

RV supernatant from an individual well was used to transduce a single well of CD4 T cells. 96 independent RV supernatants were done in parallel (**Figure 3A**). This was done to ensure that CD4 T cells were transduced with a single shRNAmir construct. The plates containing the stimulated CD4 T cells were spun down and the medium was removed and replaced with RV supernatant from the corresponding well, supplemented with 50 μ M BME and 8 μ g/mL polybrene (Millipore), followed with centrifugation for 90 min at 1500 rpm and 37°C. CD4 T cells were transduced twice: a first time at 24 hours after in vitro stimulation of CD4 T cells and the second time 12 hours later. After each transduction, the RV-containing medium was replaced with D-10 + 50 mM BME + 10 ng/ml hIL-2. 24 hours after the second transduction, cells were transferred into 48-well plates in fresh D-10 + 50 mM BME + 10 ng/ml hIL-2 and cultured for 2.5 additional days (*in vitro* expansion, **Figure 3A**). Medium was replaced with D-10 + 50 mM BME + 2 ng/ml hIL-7 for 16 hours prior to cell sorting (*in vitro* rest, **Figure 3A**). After the *in vitro* IL-7 rest, CD4 T cells were harvested and a small aliquot of each well was kept to verify the transduction efficiencies of individual shRNAmir-RV constructs. Remaining cells were combined and Amt^{hi} cells (top 50% MFI of Amt⁺ cells) were sorted from the pooled CD4 T cells. 5x10⁵ sorted Amt^{hi} cells were adoptively transferred in B6 mice (5 mice per group). From the remaining cells, DNA was isolated from 5x10⁵-1x10⁶ Amt^{hi} cells as the “Input” sample (**Figure 3A**) using the FlexiGene DNA kit (Qiagen).

Mouse infections and isolation of Tfh and Th1 CD4 T cells. Cells were “rested” for 3.5 days in B6 mice prior to infection. This allows sufficient time for the cells to return to a quiescent state (Johnston et al., 2009; McKinstry et al., 2007). Mice were infected 3.5 days after cell transfer with 2 x 10⁶ PFU of LCMV Armstrong. 6 days after infection, spleens from 5 recipient mice were combined and CD4 T cells from whole splenocytes were isolated by negative selection using magnetic beads (Miltenyi) and donor SMARTA CD4 (CD4⁺CD45.1⁺Ametrine^{hi}CD44^{hi}CD62L^{low}) were sorted using a BS FACS Aria (BD Biosciences) into Tfh and Th1 populations based on CXCR5 and SLAM expression with Tfh cells defined as SLAM^{lo}CXCR5^{high} and Th1 as SLAM^{hi}CXCR5⁻ cells (**Figure 3A**) (Johnston et al., 2009). 1.5x10⁶ Th1 and 2.5x10⁶ Tfh cells were routinely obtained after sort. DNA was isolated from each population as described above (“Tfh” and “Th1” samples).

Next generation sequencing. The shRNA_{mir} library was sequenced using the Ion Torrent technology from Life Technologies. Deep sequencing libraries were generated by PCR amplification of shRNA guide strands using primers that tag the product with standard Ion Torrent adaptors (the Ion Torrent adaptors are underlined: primerA+mir30, CCATCTCATCCCTGCGTGTCTCCGACTCAG CTCGAGAAGGTATATTGCT; primerP1+Loop, CCTCTCTATGGGCAGTCGGTGAT TACATCTGTGGCTTCACTA). All shRNA_{mir} in the library are amplified in three individual 50 μ l PCR reaction containing 100ng of genomic DNA, 1U of Phusion Hot Start II High-Fidelity DNA polymerase (Thermo Scientific), 1x Phusion HF buffer, 0.5 μ M of each primer and 0.2 μ M of each dNTPs using the following cycling parameters: 98 °C for 2 min; 26 cycles of 98 °C for 30 s, 58 °C for 30 s and 72 °C for 20 s; 72 °C for 2 min. The shRNA library PCR amplicon was then cleaned using Agencourt AMPure XP beads (Beckman Coulter). The quality of the library after cleanup was assessed using the Agilent High Sensitivity DNA Kit and the Agilent 2100 Bioanalyzer (Agilent Technologies). The library was diluted and used to prepare template-positive Ion Sphere Particles (ISPs) with the Ion OneTouch 200 Template Kit v2 and the Ion One Touch instrument (Life Technologies). The template-positive ISPs were enriched with the Ion OneTouch 200 Template Kit v2 and the Ion OneTouch ES (Life Technologies). The enriched template-positive ISPs were loaded onto an Ion 314 Chip, sequenced with the Ion PGM 200 Sequencing Kit and the Personal Genome Machine System (Life Technologies). 250,000-600,000 reads were routinely obtained, per sample. In some experiment the higher density Ion 318 Chips were used.

Sequencing analysis. The sequencing data was first analyzed by using BLAST with default mismatch settings, to align the sequencing reads to a reference library. To generate the reference library, the sequences ACAGTGAGCG and TAGTGAAGCCACAGATGTA were added to the 5' and 3' ends of the shRNA sense strand, respectively, of each clone.

For CD8 T cell experiments: the total number of reads after sequencing shRNA_{mir}s in the different T cell subsets was scaled to the input sample, and the number of reads for each individual shRNA_{mir} was normalized accordingly. To provide a sufficient baseline to detect possible shRNA depletion in differentiated T cell samples, shRNAs with read counts lower than 50 in the Input were excluded from the analysis. Samples with read counts lower than 50 in the differentiated T cell samples were considered for the analysis if their number of reads was higher than 50 in the Input. Some samples were not detected in the differentiated T cell samples and were considered as depleted. Their read counts were set at one for the analysis. To quantify enrichment of shRNA_{mir}s in the CTL subsets, the

read number for each shRNA_{mir} in memory precursor CTL was divided by the corresponding shRNA_{mir} read number in effector CTL, and the resulting values were transformed to $\log_2$. The average and standard deviation of the ratios of shRNA_{mir}s targeting negative control genes (*Cd4*, *Cd14*, *Cd19*, *Ms4a1* (CD20)) was determined and these values were used to calculate Z-scores for each shRNA_{mir} and were plotted (**Figure 2A**). In a separate analysis, the normalized effector or memory precursor CTL reads were divided by the Input reads and the $\log_2$ values for all ratios were calculated and plotted against each other (**Figure 2B**).

For CD4 T cell experiments: the number of reads for each shRNA_{mir} was normalized to its number of reads in the Input sample. Normalized Tfh reads were divided by the normalized Th1 reads to obtain Tfh/Th1 ratios. The $\log_2$ of these ratios was calculated. Conventional Z-score values for every shRNA ratio were calculated using the mean and standard deviation of the negative control set of shRNAs (*Cd14*, *Cd19*, *Cd22*, *Ms4a1* (CD20), *Cd8*, *Smarca1*) (**Figure 3B**). In a separate analysis, the normalized Tfh or Th1 reads were divided by the Input reads and the $\log_2$ values for all ratios were calculated and plotted against each other (**Figure 3C**). To provide a sufficient baseline to detect possible shRNA depletion in differentiated T cell samples, samples with read counts lower than 50 in the Input were excluded from the analysis. Samples with read counts lower than 50 in the differentiated T cell samples were considered for the analysis if their number of reads was higher than 50 in the Input. Some samples were not detected in the differentiated T cell samples and were considered as depleted. Their read counts were set at one for the analysis. A Z score of $\geq |3.0|$ for at least one shRNA_{mir} against a gene of interest was the primary selection criterion for further consideration. If shRNA_{mir}s against a gene gave conflicting outcomes (e.g., Tfh bias for one, Th1 bias for another), the gene was discarded.

Supplemental Detailed Protocols

Detailed CD8 protocol

Materials

- Platinum-E (PLAT-E) cells
- 150-mm tissue culture dishes (Fisher, cat. # 0877224)
- DMEM (Fisher, cat. # MT10017CV)
- Fetal bovine serum (Omega scientific, cat. # FB-12)
- Penicillin/Streptomycin (Life Technologies, cat. # 15140163)
- Glutamax (L-glutamine, Life Technologies, cat. # 35050079)
- HEPES (1M) (Life Technologies, cat. # 15630-080)
- 2-mercaptoethanol (BME)
- 0.25% Trypsin/EDTA (Life Technologies, cat. # 25200114)
- PBS (Fisher, cat. # MT21040CV)
- Bovine serum albumin (Sigma Aldrich, cat. # A7906)
- UltraPure 0.5M EDTA, pH 8.0 (Life Technologies, cat. # 15575-020)
- Sodium azide (Fisher, cat. # BP922I-500)
- TransIT-LT1 (Mirus, Cat.# MIR 2300)
- Opti-MEM® I Reduced Serum Medium (Life Technologies, cat. # 31985-070)
- pCI-Eco (Addgene, cat. #12371)
- Anti-CD3 antibody (Homemade, clone 2C11)
- Anti-CD28 antibody (BD-Pharm, clone 37.51, cat. # 553295)
- Goat Anti-Hamster IgG (H+L) (ThermoScientific, cat. # 31115)
- Recombinant human IL-2 (PeproTech, cat. # 200-02)
- 70-µm cell strainer (Fisher, cat. # 22363548)
- 1 mL SlipTip Sterile Syringe (without needle) (Fisher, cat. # 14823434)
- ACK Lysing buffer (Life Technologies, cat. # A10492-01)
- 1 mL syringe (with 27 G needles) (Fisher, cat. # 14823434)
- Dynal Mouse CD8 Negative Isolation Kit (Invitrogen, cat. # 114.17D)
- CD90.1 MicroBeads (Miltenyi, cat. # 130-094-523)
- QuadroMACS Separator (Miltenyi, cat. # 130-090-976) and MACS MultiStand (Miltenyi, cat. # 130-042-303)
- MACS LS Columns (Miltenyi, cat. # 130-042-401)
- Polybrene (Millipore, cat. # TR-1003-G)
- Tissue culture-treated 96 well flat plate (Fisher, cat. # 087721G)
- Sterile 96 well round bottom plate (Fisher, cat. # 07200760)
- 15 mL conical tube (Fisher, cat. # 1495949B)
- 50 mL conical tube (Fisher, cat. # 1443222)
- CD45.1⁺ P14 TCR transgenic mice (P14 mice can be purchased from Taconic Farms, cat. #4138)
- C57Bl/6 mice (The Jackson Laboratory (Maine), cat. # 000664)
- LCMV clone13
- FlexiGene DNA kit (Qiagen cat. # 51204)
- Anti-KLRG-1 APC (BioLegend, cat. # 138412)

- Anti- IL-7R α PE (BioLegend, cat. # 135010)
- Anti-CD8 PerCP-Cy5.5 (eBioscience, cat. # 45-0081)
- Anti-CD90.1 PE-Cy7 (BioLegend, cat. # 202518)
- Anti-CD4 APC-eFluor780 (eBioscience, cat. # 47-0042)
- Anti-B220 APC-eFluor780 (eBioscience, cat. # 47-0452)
- Ethanol (Fisher, Cat. # BP28184)
- Isopropanol (Fisher, Cat. # BP26324)

Reagent setup

- Complete medium (PLAT-E medium): 500 mL DMEM + 10% FBS + 10 mM HEPES + 50 U/mL Penicillin/Streptomycin + 2 mM Glutamax+ 50 μ M BME
- cTCM (complete T-cell medium): 500 mL DMEM + 10% FBS + 10 mM HEPES + 50 U/mL Penicillin/Streptomycin + 2 mM Glutamax+ 50 μ M $\square\square\square\square$
- MACS buffer: 500 mL PBS + 0.5% bovine serum albumin + 2 mM EDTA
- FACS buffer: 500 mL PBS + 1% Fetal bovine serum

Equipment

- Flow cytometer: BD Biosciences or any that permits use of Ametrine (violet laser, 505LP filter)
- Fluorescence-activated cell sorting (FACS) machine: BD Biosciences FACSria or any that permits use of Ametrine (violet laser, 505LP filter)

Production of retroviruses (RV) (days 1-2)

1. Plate PLAT-E in 96 well plates in the afternoon. Seed 6×10^4 cells per well in 100 μ L of complete medium without Penicillin/Streptomycin. Avoid the edges of the plates by plating cells only in the central 60 wells. Fill the remaining wells with 100 μ L of medium.
2. Proceed to transfection of PLAT-E on the following afternoon.
3. Change medium 5 hrs before transfection. Remove medium from PLAT-E and carefully replace with 100 μ L of pre-warmed (37°C) complete medium without Penicillin/Streptomycin.
4. Warm TransIT-LT1 Reagent and Opti-MEM® I Reduced Serum Medium to room temperature and vortex gently before using.
5. Prepare transfection mix in round-bottom 96 well plates.
6. Add 0.2 μ g shRNAmir vector + 0.2 μ g pCL-Eco vector to 10 μ L Opti-MEM® I Reduced Serum Medium, pipet gently to mix completely.
7. Add 0.6 μ L TransIT-LT1 to 10 μ L Opti-MEM® I Reduced Serum Medium, then immediately add 10 μ L TransIT-LT1/Opti-MEM® I Reduced Serum Medium to the diluted DNA mixture, pipet gently to mix completely, incubate at room temperature for 20 min.
8. Add the TransIT-LT1 Reagent/DNA complexes drop-wise to different areas of the wells. Gently swirl the plate to mix and incubate at 37°C overnight.
9. 16 hours later, remove medium from PLAT-E and carefully replace with 100 μ L of pre-warmed (37°C) cTCM.

P14 CD8 T cell stimulation (day 4)

10. Coat 96 well plates with 50 μ L of 100 μ g/mL Goat Anti-Hamster IgG (H+L) diluted in PBS (avoid edges of the plates as in step 1). Incubate plates at room temperature for 5 hrs.
11. Isolate P14 CD8 T cells from pooled spleens of CD45.1⁺P14 TCR Tg mice (2 spleens are required to obtain enough CD8 T cells for 96 wells) using the Dynal Mouse CD8 Negative Isolation Kit.
 - a. Harvest spleen and transfer to 15 mL tube with 2 mL cTCM
 - b. Pour up to 2 spleens into 70 μ m filter on top of a 50 mL tube. Mash spleens through filter with the thumb end of a 1mL syringe plunger.
 - c. Rinse once with 8 mL cTCM, mash again, and then rinse with 10 mL cTCM.

- d. Harvest cells at 300 x g for 10 min at 4°C.
- e. Resuspend cells in 1 mL ice cold 1x PBS/1.0% FBS and transfer to an eppendorf tube.
- f. Combine cells from 1 spleen and the depletion antibody cocktail in the following order and ratios:
 - 1,000 µL cells
 - 200 µL FBS
 - 100 µL depletion antibody cocktail
- g. Incubate cells, rotating with antibody cocktail for 20 min at 4°C. Pre-wash Dynal beads three times with ice cold PBS/1.0%FBS during incubation [use 1 mL beads per 10⁸ input cells]. Resuspend Dynal beads in vial and transfer required volume to an appropriately sized tube.
- h. After incubation, transfer cells/depletion antibody mixture to a fresh 15 mL conical, wash cells by adding up to 15 mL with ice cold PBS/1.0%FBS, rinse eppendorf with 1 mL PBS/1.0%FBS and pool. Harvest cells at 300 x g for 10 min at 4°C, aspirate supernatant, resuspend cell pellet in 7 mL PBS/1.0%FBS in a 15 mL conical tube.
- i. Add pre-washed magnetic beads to resuspended cells; incubate rotating at room temperature for 15 min.
- j. After incubation, keep inverting tube and immediately apply to magnet, leave tube on magnet ~2 min.
- k. With tube remaining on magnet, collect the supernatant by pipetting- transfer to a fresh 15 mL conical tube, add 8 mL fresh ice cold cTCM. Invert tube on magnet to rinse beads/tube.
- l. Collect the rinsing volume and pool with first fraction of cells.
- m. Harvest cells at 300 x g for 10 min at 4°C.
- n. Resuspend cells in 1 mL fresh cTCM and transfer to a 4 mL round bottom tube and place this tube on magnet for 2 min to eliminate any residual beads that will be present.
- o. Recover the cells and transfer to a fresh tube and rinse tube on magnet 1x with fresh cTCM.
- p. Count cells. Adjust cell concentration to 2x10⁶ cells/mL in cTCM.
- q. Prepare the activation medium (2 µg/mL of anti-CD3 and anti-CD28 in cTCM).
- r. Rinse pre-coated plates twice with room temperature cTCM before plating cells
- s. Combine equal volumes of CD8 T cells and the activation medium; mix and immediately dispense 200 µL into each Goat Anti-Hamster IgG (H+L) pre-coated wells. Avoid the edges of the plates as in step 1. Fill in the used wells with warm cTCM.
12. Harvest retrovirus (RV)-containing supernatant from PLAT-E in the morning and afternoon. Pipet 70 µL of the medium from the PLAT-E into fresh 96 well plates. Store the plates containing the RV at 4°C. Gently add 70 µL of fresh complete medium to the PLAT-E.

Transduction (day 5)

13. The optimal time for the first transduction is 18 hours post stimulation.
14. Pre-warm the centrifuge to 37°C.
15. Harvest the RV-containing supernatant as in step 12 and mix with the corresponding supernatant stored at 4°C.
16. Add 2-mercaptoethanol to a final concentration of 50 µM and 8 µg/mL polybrene to RV supernatant and mix.
17. Carefully add RV mixes to the CD8 T cells in the 96 well plates.
18. Spin at 2,000 rpm for 1 hour at 37°C.
19. Transfer 96 well plates to incubator (37°C 10% CO₂) for an additional 4 hrs.
20. Remove 200 µL of viral supernatants from each well and replace with 200 µL of cTCM and culture at 37°C 10% CO₂ until 48 hrs post activation.

Adoptive transfer and LCMV infection (day 6)

21. 48 hrs post activation, pool cells, determine cell concentration and spin down for 5 minutes at 1500 rpm before adoptive transfer.
22. Discard supernatant and resuspend cells with plain DMEM in 2.5×10^6 cells/mL and 200 μ L of the cell solution (corresponding to 5×10^5 cells) is injected into C57Bl/6 recipient mice by the retro-orbital route using an insulin needle.
23. To reduce variation in individual mice and increasing the cell number available for the following procedures, cells are injected into 8 recipient mice.
24. Sort the remaining cells based on the top 20% Ametrine1.1 MFI of Amt⁺ cells as the "Input" sample. Genomic DNA was isolated from 3×10^5 sorted cells for the "input" library. (refer to steps 34-47).
25. 1 hour after adoptive transfer, infect transferred mice with LCMV-clone13.
26. Prepare a solution of LCMV-clone13 at a concentration of 2×10^5 PFU/mL in plain DMEM.
27. Inject 500 μ L of the LCMV-clone13 solution intra-peritoneally per mouse with a 1 mL syringe. Inject 250 μ L on each side of the mouse.

Sort effector and memory precursor populations after LCMV infection (day 13)

28. 7 days after LCMV infection, mice are euthanized and spleens harvested. Spleens from 8 recipient mice are pooled at this step.
29. Total CD8 T cells are isolated with CD90.1 MicroBeads from Miltenyi.
 - a. Harvest spleen and transfer to 15 mL tube with 2 mL cTCM
 - b. Pour spleen into 70 μ M filter on top of a 50 mL tube. Mash spleens through filter with the thumb end of a 1 mL syringe plunger.
 - c. Rinse once with 8 mL cTCM, mash again, and then rinse with 10 mL cTCM.
 - d. Harvest cells at 300 x g for 10 min at 4°C.
 - e. Discard supernatant and resuspend pellet with 2 mL ACK Lysing buffer and incubate at room temperature for 3 min.
 - f. Add 8 mL cTCM, mix and take an aliquot to determine cell concentration.
 - g. Centrifuge cell suspension at 300 x g for 10 min at 4°C. Aspirate supernatant completely.
 - h. Resuspend cell pellet in 90 μ L of MACS buffer per 10^7 total cells.
 - i. Add 10 μ L of CD90.1 MicroBeads per 10^7 total cells.
 - j. Mix well and incubate for 15 minutes at 4°C.
 - k. Wash cells by adding 2 mL of MACS buffer per 10^7 cells and centrifuge at 300 x g for 10 min at 4°C. Aspirate supernatant completely.
 - l. Resuspend up to 10^8 cells in 500 μ L of MACS buffer.
 - m. Place column in the magnetic field of a suitable MACS Separator.
 - n. Prepare column by rinsing with 3 mL of MACS buffer.
 - o. Apply cell suspension onto the column. Collect flow-through containing unlabeled cells.
 - p. Wash column with 3 mL of MACS buffer. Collect unlabeled cells that pass through and combine with the effluent from step o.
 - q. Remove column from the separator and place it on 15 mL tube.
 - r. Pipette 5 mL of MACS buffer onto the column. Immediately flush out the magnetically labeled cells by firmly pushing the plunger into the column.
30. Isolated CD8 are stained with the following antibody panel prior to sort
 - a. PE: IL-7R α
 - b. PerCP-Cy5.5: CD8
 - c. PE-Cy7: CD90.1
 - d. APC: KLRG-1

- e. APC-eFluor780: CD4⁻; B220
- 31. After staining, wash cells twice with FACS buffer and resuspend cells in FACS buffer at a concentration appropriate for cell sorting.
- 32. Sort P14 effector and memory precursor populations based on the top 20% Ametrine^{1.1} MFI of Amt⁺ cells
 - a. P14 effector: CD4⁻ B220⁻ CD8⁺ CD90.1⁺ Ametrine⁺ KLRG-1⁺ IL-7Rα⁻
 - b. P14 memory precursor: CD4⁻ B220⁻ CD8⁺ CD90.1⁺ Ametrine⁺ KLRG-1⁻ IL-7Rα⁺
- 33. Fill tubes of sorted P14 effector and memory precursor with PBS and spin down 10 minutes at 1500 rpm. Discard supernatant.

Isolation of genomic DNA

- 34. Genomic DNA is isolated with the FlexiGene DNA kit, following a modified protocol for the isolation of DNA from 1-2x10⁶ cultured cells.
- 35. Add 300 µL Buffer FG1 to the cell pellet and mix by pipetting until the cells are resuspended.
- 36. Centrifuge for 1 min at 10,000 x g in a fixed-angle rotor.
- 37. Discard the supernatant and leave the tube inverted on a clean piece of absorbent paper for 2 min, taking care that the pellet remains in the tube.
- 38. Add 300 µl Buffer FG2/QIAGEN Protease, close the tube, and vortex immediately until the pellet is completely homogenized. Inspect the tube to check that homogenization is complete.
- 39. Invert the tube 3 times, place it in water bath, and incubate at 65°C for 5 min. Add 300 µl isopropanol (100%) and mix thoroughly by inversion until the DNA precipitate becomes visible as threads or a clump.
- 40. Centrifuge for 3 min at 10,000 x g.
- 41. Discard the supernatant and briefly invert the tube onto a clean piece of absorbent paper, taking care that the pellet remains in the tube.
- 42. Add 300 µl 70% ethanol and vortex for 5 sec.
- 43. Centrifuge for 3 min at 10,000 x g.
- 44. Discard the supernatant and leave the tube inverted on a clean piece of absorbent paper for at least 5 min, taking care that the pellet remains in the tube.
- 45. Air dry the DNA pellet for 5 min until all the liquid has evaporated.
- 46. Add 200 µl Buffer FG3, vortex for 5 sec at low speed, and dissolve the DNA by incubating for 5 min at 65°C for at least 1 hour.
- 47. Store the isolated genomic DNA at -20°C until further use.

Detailed CD4 protocol

Materials

- Platinum-E (PLAT-E) cells
- 150-mm tissue culture dishes (Fisher, cat. # 0877224)
- DMEM (Fisher, cat. # MT10017CV)
- Fetal bovine serum (Omega scientific, cat. # FB-12)
- Penicillin/Streptomycin (Life Technologies, cat. # 15140163)
- Glutamax (L-glutamine, Life Technologies, cat. # 35050079)
- 2-mercaptoethanol (BME) (Sigma-Aldrich, cat. # M3148)
- 0.25% Trypsin/EDTA (Life Technologies, cat. # 25200114)
- PBS (Fisher, cat. # MT21040CV)
- Bovine serum albumin (Sigma Aldrich, cat. # A7906)
- UltraPure 0.5 M EDTA, pH 8.0 (Life Technologies, cat. # 15575-020)
- Sodium azide (Fisher, cat. # BP922I-500)
- Polyethylenimine, branched (PEI) (Sigma-Aldrich, cat. # 408727)
- pCL-Eco (Addgene, cat. #12371)
- Anti-CD3e antibody (BioXCell, clone 145-2C11, cat. # BE0001-1)
- Anti-CD28 antibody (BioXCell, clone 37.51, cat. # BE0015-1)
- Recombinant mouse IL-7 (PeproTech, cat. # 217-17)
- Recombinant human IL-2 (PeproTech, cat. # 200-02)
- 70-µm cell strainer (Fisher, cat. # 22363548)
- 1 mL SlipTip Sterile Syringe (without needle) (Fisher, cat. # 14823434)
- ACK Lysing buffer (Life Technologies, cat. # A10492-01)
- 1 mL syringe (with 27 G needles) (Fisher, cat. # 14823434)
- CD4 isolation kit II (Miltenyi, cat. # 130-095-248)
- QuadroMACS Separator (Miltenyi, cat. # 130-090-976) and MACS MultiStand (Miltenyi, cat. # 130-042-303)
- MACS LS Columns (Miltenyi, cat. # 130-042-401)
- Polybrene (Millipore, cat. # TR-1003-G)
- Insulin needles (U-100) (Fisher, cat. # 1482679)
- Tissue culture-treated 24 well flat plate (Fisher, cat. # 7200740)
- Tissue culture-treated 96 well flat plate (Fisher, cat. # 087721G)
- Sterile 96 well round bottom plate (Fisher, cat. # 07200760)
- 15 mL conical tube (Fisher, cat. # 1495949B)
- 50 mL conical tube (Fisher, cat. # 1443222)
- 1.7 mL Posi-Click Eppendorf Tube (Denville, cat. #C2170)
- SMARTA mice (LCMV gp66-77-IA^b specific)
- B6.SJL mice (The Jackson Laboratories (Maine), cat #002014)
- C57Bl/6 mice (The Jackson Laboratory (Maine), cat. # 000664)
- LCMV Armstrong
- FlexiGene DNA kit (Qiagen cat. # 51204)
- Anti-CD45.1 FITC (eBioscience, cat. # 11-0453)
- Anti-SLAM PE (BioLegend, cat. # 115903)
- Anti-CD8 PerCP-Cy5.5 (eBioscience, cat. # 45-0081)

- Anti-B220 PerCP-Cy5.5 (eBioscience, cat. # 45-0452)
- Anti-CD62L PE-Cy7 (eBioscience, cat. # 25-0621)
- Anti-CD44 eFluor450 (eBioscience, cat. # 48-0441)
- Anti-CD4 APC-eFluor780 (eBioscience, cat. # 47-0042)
- Rat anti-mouse CXCR5 (BD Biosciences, cat. # 551961)
- Biotin conjugated AffiniPure Goat anti-rat (H+L) (Jackson ImmunoResearch, cat. # 112-065-167)
- Streptavidin APC (eBioscience, cat. # 17-4317)
- V α 2 $\square\square\square\square$ (eBioscience, cat. # 17-5812)
- CD4 PE (eBioscience, cat. # 12-0041)
- Normal mouse serum (Sigma-Aldrich, cat. # M5905)
- Ethanol (Sigma-Aldrich, cat. # E7023)

Reagent setup

- D10: 500 mL DMEM + 10% FBS (50 mL) + 5 mL Penicillin/Streptomycin + 5 mL Glutamax
- MACS buffer: 500 mL PBS + 0.5% bovine serum albumin + 2 mM EDTA
- FACS buffer: 500 mL PBS + 0.5% bovine serum albumin + 0.1% sodium azide
- Breed SMARTA mice with B6.SJL mice to obtain SMARTA CD45.1^{+/+} mice

Equipment

- Flow cytometer: LSRII from BD Biosciences or any that permits use of Ametrine (violet laser, 505LP dichroic filter)
- Fluorescence-activated cell sorting (FACS) machine: BD Biosciences FACSria or any that permits use of Ametrine (violet laser, 505LP dichroic filter)

Production of retroviruses (RV) (days 1-2)

1. Plate PLAT-E in 96 well plates in the afternoon (~4-5 pm). Seed 4×10^4 cells per well in 200 μ L of D10. Avoid the edges of the plates by plating cells only in the central 60 wells. Fill the remaining wells with 200 μ L of medium.
2. Proceed to transfection of PLAT-E on the following morning (~8-9 am).
3. Change medium 1 hour before transfection. Remove medium from PLAT-E and carefully replace with 100 μ L of pre-warmed (37°C) D10. Pre-warm plain DMEM to room temperature for Step 5.
4. 30 minutes later, prepare transfection mix in round-bottom 96 well plates.
5. Mix 0.5 μ L PEI with 10 μ L plain DMEM (pre-warmed to RT), incubate at RT for 10 minutes.
6. Add 0.1 μ g shRNA_{mir} vector + 0.2 μ g pCL-Eco vector to PEI/DMEM solution, mix gently and incubate 15 minutes at RT. NOTE: It is more convenient to prepare a plate of the shRNA_{mir} vectors with all wells having the same DNA concentration.
7. Gently add DNA/PEI/DMEM solution to the wells, drop by drop. Gently swirl the plate to mix and incubate at 37°C for 6 hours. NOTE: You can work with a multi-channel pipette for steps 5-7.
8. 6 hours later, gently add 100 μ L of fresh warm D10 to every well by pipetting down on the sides of the wells.

SMARTA CD4 T cell stimulation (day 3)

9. In the afternoon of the following day (~2-3 pm), isolate CD4 T cells and plate in anti-CD3/CD28 coated wells
10. Coat 96 well plates with 100 μ L 8 μ g/mL anti-CD3e and anti-CD28 diluted in PBS (avoid edges of the plates as in step 1 by keeping the same plate layout used for the PLAT-E). Incubate plates at 37°C for at least 1 hour.
11. Isolate CD45.1⁺ SMARTA CD4 T cells from pooled spleens of SMARTA CD45.1^{+/+} mice (2 spleens are required to obtain enough CD4 T cells for 96 wells) using the Miltenyi CD4 T cell isolation kit II.

- a. Crush the spleens on a 70- μ m cell strainer over a 50 mL conical tube with the plunger from a 1 mL sterile syringe. Rinse the strainer twice with 10-13 mL D10.
- b. Spin down cells 5 min at 1600 rpm and 4°C.
- c. Discard supernatant and resuspend pellet with 4 mL ACK Lysing buffer (2 mL/spleen) and incubate at room temperature for 2 min.
- d. Add 16 mL D10 (8 mL/spleen), mix and take an aliquot to determine cell concentration.
- e. Spin down cells as in step 11b.
- f. Discard supernatant. Resuspend pellet in 40 μ L MACS buffer per 10^7 cells. Filter the cells through a 70- μ m cell strainer into a new 50 mL conical tube. Add 10 μ L antibody cocktail per 10^7 cells. Incubate 10 min at 4°C, shaking every 5 minutes.
 - i. During the incubation, prepare D10 + 50 μ M BME + 2 ng/mL mIL-7 and pre-warm at 37°C. Calculate 200 μ L per well to plate.
- g. Add 30 μ L MACS buffer per 10^7 cells. Add 20 μ L biotin beads per 10^7 cells. Incubate 15 min at 4°C, shaking every 5 minutes.
- h. Prepare MACS LS columns
 - i. Determine the number of columns to be used. Each column can be loaded with up to 1×10^8 magnetically labeled cells. Assume that ~75% of the cells from a spleen of a SMARTA TCR Tg mouse are labeled with the CD4 T cell isolation kit II.
 - ii. Place LS Column on QuadroMacs Separator placed on the MACS MultiStand. Place a 15 mL conical tube under the column to collect the flow-through.
 - iii. Wash column with 6 mL MACS buffer. Prepare the column at the last minute to keep it cold and prevent it from drying.
- i. Add 1-2 mL MACS buffer per 10^7 cells and spin down as in step 11b.
- j. Resuspend cells in MACS buffer: up to 10^8 cells in 500 μ L.
- k. Load cells onto column and collect flow-through.
- l. Wash column with 5 mL MACS buffer and collect flow-through.
- m. Take an aliquot to verify purity of isolated CD4
 - i. Stain with the following antibodies
 1. FITC: CD45.1
 2. PE: CD4
 3. PerCP-Cy5.5: CD8, B220
 4. eFluor450: CD44
 5. APC: V α 2
 - ii. The isolated cells should be CD8⁻ B220⁻ CD4⁺ CD45.1⁺ V α 2⁺ and mostly CD44⁻ (~85-90%).
- n. Spin down cells as in step 11b.
- o. Resuspend cells in 1 mL pre-warmed D10 + 50 μ M BME + 2 ng/mL mIL-7. Take an aliquot and determine cell concentration.
- p. Adjust cell concentration to 1×10^6 cells/mL in D10 + 50 μ M BME + 2 ng/mL mIL-7.
12. Remove PBS from the wells of the 96 well plates by flicking.
13. Seed 2×10^5 SMARTA cells per well by pipetting 200 μ L in each well coated with anti-CD3e and anti-CD28 antibodies. Use a multichannel pipette to work faster and to avoid the wells from drying. Fill in the unused wells with warm D10. It is critical to gently pipette the cells in the wells to obtain a uniform distribution of the cells resulting in efficient activation.
14. Harvest retrovirus (RV)-containing supernatants from PLAT-E. Pipet 170 μ L of medium from the PLAT-E into fresh 96 well plates for storage at 4°C. Gently add 180 μ L of warm fresh D10 to the PLAT-E by pipetting down on the sides of the wells.

1st transduction (day 4)

15. The optimal time for the first transduction is 22-24 hours post stimulation to time with the mitotic phase (~3-4 pm)
16. Pre-warm the centrifuge to 37°C.
17. Harvest the RV-containing supernatants as in step 14 and mix with the corresponding supernatants stored at 4°C.
18. Transfer 150 μ L of the RV supernatants to fresh 96 well plates to prepare the RV transduction mixes. Keep the remaining RV supernatant at 4°C for the second transduction.
19. Supplement RV supernatants with 50 μ M BME and 8 μ g/ml polybrene and mix by pipetting up and down.
20. Spin down the plates of SMARTA CD4 for 1 min at 1500 rpm. Remove medium.
21. Carefully add RV mixes to the CD4 on the 96 well plates by pipetting down on the sides of the wells.
22. Spin for 2 hours at 1500 rpm and 37°C.
23. Prepare D10 + 50 μ M BME + 10 ng/mL hIL-2 and pre-warm at 37°C. Calculate 200 μ L
24. Remove supernatant and gently replace with pre-warmed D10 + 50 μ M BME + 10 ng/mL recombinant hIL-2.
25. Verify transfection efficiency of all wells of PLAT-E by flow cytometry.
 - a. Add 100 μ L of FACS buffer to all wells and pipet each well separately into a tube for acquisition on a flow cytometer.

2nd transduction (day 5)

26. Proceed with the second transduction approximately 36 hours post-stimulation (~8 am).
27. Repeat steps 16, 18-25.

IL-2 expansion (day 6)

28. Separately move transduced SMARTA to 24 well plates for expansion 24 hours after the second transduction (~9-10 am).
29. Pre-warm D10 + 50 μ M BME + 10 ng/mL hIL-2. Calculate 1 mL per well.
30. Pipet CD4 from each well of the 96 well plates to individual wells of 24 well plates (keep CD4 T cells separated).
31. Add 1 mL of warm fresh D10 + 50 μ M BME + 10 ng/mL hIL-2 to each well.

IL-7 *in vitro* rest (day 8)

32. Cells are given fresh D10 + 50 μ M BME + 2 ng/mL mL-7 16-18 hours prior to transfer (~5-6 pm).
33. Pre-warm D10 + 50 μ M BME + 2 ng/mL mL-7. Calculate 1 mL per well.
34. Spin down plates of SMARTA CD4 for 1 min at 1500 rpm. Carefully remove medium.
35. Add 1 mL of warm D10 + 50 μ M BME + 2 ng/mL mL-7.

Sort and adoptive transfer (day 9)

36. Cells are pooled before sorting the transduced cells for adoptive transfer (~8-9 am).
37. Harvest a small aliquot (~25 μ L) of cells from each well to verify the transduction efficiency of every shRNAmir construct.
38. Harvest and pool SMARTA from all the wells and verify the transduction efficiency of the pooled cells.
39. Sort cells expressing high levels of Ametrine (top 50%).
40. After sort, the cells are prepared for adoptive transfer. The tube containing the Ametrine-high cells is filled with PBS and the cells are spun down for 10 minutes at 1500 rpm at 4.
41. Discard supernatant and wash cells with plain DMEM; spin cells as in step 40.
42. Discard supernatant and resuspend cells in 1 mL plain DMEM. Determine cell concentration.

43. Cells are diluted to 5×10^6 cells/mL in plain DMEM and 100 μ L of the cell solution (corresponding to 5×10^5 cells) is injected into C57Bl/6 recipient mice by the retro-orbital route using an insulin needle.
44. To reduce variation caused by variable engraftment in individual mice and to increase the cell number available for the following procedures, cells are injected into 5 recipient mice.
45. Cells are allowed to rest *in vivo* for at least 3.5 days before infection.
46. Genomic DNA is isolated from 1×10^6 sorted cells for the “input” library. (refer to steps 56-67).

LCMV infection (day 13)

47. Mice are infected with LCMV-Armstrong (~8-9 am).
48. Prepare a solution of LCMV-Armstrong at a concentration of 4×10^6 PFU/mL in plain DMEM.
49. Inject 500 μ L of the LCMV-Armstrong solution (corresponding to 2×10^6 PFU) intra-peritoneally per mouse using a 1 mL syringe (with 27 G needle). Inject 250 μ L on each side of the mouse.

Sort Tfh and Th1 populations after LCMV infection (day 19)

50. 6 days after LCMV infection, mice are euthanized (~8 am) and spleens harvested. Spleens from 5 recipient mice are pooled at this step.
51. Total CD4 T cells are isolated with a CD4 isolation kit II from Miltenyi as described in step 11.
52. Isolated CD4 are stained with the following antibody panel prior to sort (stain cells at a concentration of 2.5×10^7 cells/mL)
 - b. FITC: CD45.1
 - c. PE: SLAM
 - d. PerCP-Cy5.5: CD8, B220
 - e. PE-Cy7: CD62L
 - f. eFluor450: CD44
 - g. Ametrine
 - h. APC: CXCR5 (3 steps staining protocol, detailed below)
 - i. APC-eFluor780: CD4
53. CXCR5 3 steps staining protocol
 - a. Resuspend isolated CD4 at a concentration of 1.5×10^7 cells/mL in FACS buffer + serum (2% normal mouse serum + 2% FBS). Add 10 μ L/mL (1:100 dilution) of primary CXCR5 antibody (rat anti-mouse CXCR5 from BD). Incubate 1 hour at 4°C, shaking tube every 15 minutes.
 - b. Wash twice with FACS buffer by filling the tube with FACS buffer and spinning down 5 min and 4°C.
 - c. Resuspend isolated CD4 at a concentration of 1.5×10^7 cells/mL in FACS buffer + serum (2% normal mouse serum + 2% FBS). Add 1 μ L/mL (1:1000 dilution) of secondary antibody (Jackson Immunoresearch biotin conjugated AffiniPure Goat anti-rat (H+L)). Incubate 30 minutes at 4°C, shaking tube every 15 minutes.
 - d. Wash twice with FACS buffer as in step 53b.
 - e. Resuspend isolated CD4 at a concentration of 1.5×10^7 cells/mL in FACS buffer. Stain with streptavidin-APC (1:1000) and antibody cocktail detailed in step 52 for 30 minutes at 4°C, shaking tube every 15 minutes.
 - f. Wash twice with FACS buffer as in step 53b and resuspend cells in FACS buffer at a concentration appropriate for cell sorting ($1-2 \times 10^7$ cells/mL).
54. Sort SMARTA Tfh and Th1 populations
 - a. SMARTA Tfh: CD8⁻ B220⁻ CD4⁺ CD45.1⁺ Ametrine⁺ CD44⁺ CD62L⁻ CXCR5⁺ SLAM^{lo}
 - b. SMARTA Th1: CD8⁻ B220⁻ CD4⁺ CD45.1⁺ Ametrine⁺ CD44⁺ CD62L⁻ SLAM^{high} CXCR5⁻
55. Fill tubes of sorted SMARTA Tfh and SMARTA Th1 with PBS and spin down 10 minutes at 1500 rpm at 4°C. Discard supernatant.

Isolation of genomic DNA

56. Genomic DNA is isolated with the FlexiGene DNA kit from Qiagen following a modified protocol for the isolation of DNA from $1-2 \times 10^6$ cultured cells.
57. Add 300 μ L Buffer FG1 to the cell pellet and mix by pipetting until the cells are resuspended. Transfer to a 1.7 mL Eppendorf tube.
58. Prepare Buffer FG2 + Qiagen protease that has been previously resuspended in buffer FG3 as per kit's instructions.
59. Add 300 μ L Buffer FG2 + 3 μ L Qiagen protease, close the tube and invert 3 times. Incubate at 65°C for 10 minutes.
60. Add 600 μ L isopropanol (100%) and mix thoroughly by inversion until the DNA precipitate becomes visible.
61. Centrifuge for 10 minutes at top speed in a table top centrifuge. Discard the supernatant and briefly invert the tube onto a piece of clean absorbent paper.
62. Add 600 μ L 70% ethanol and vortex for 5 seconds.
63. Centrifuge for 5 minutes at top speed.
64. Discard the supernatant and leave the tube inverted on a clean piece of absorbent paper for 5 minutes.
65. Air-dry the pellet for 5 minutes.
66. Add 300 μ L Buffer FG3, vortex for 5 seconds and dissolve the pellet by incubating at 65°C for at least 1 hour.
67. Store the isolated genomic DNA at -20°C until further use.

Detailed CD8 deep sequencing and analysis

Materials

- Qubit dsDNA HS Assay Kit (Life Technologies, cat. # Q32854)
- Qubit Fluorometer (Life Technologies)
- PCR machine
- Agilent 2100 Bioanalyzer (Agilent Technologies)
- High Sensitivity DNA analysis kit (Agilent Technologies, cat. #5067-4626)
- Deep sequencing machine
 - o We use the Ion OneTouch 2 System from Life Technologies to prepare the sequencing libraries with the following kits:
 - Ion OneTouch 200 Template Kit v2 DL (Life Technologies, cat. #4480285)
 - Ion Sphere Quality Control Kit (Life Technologies, cat. #4468656)
 - o We use the Ion PGM System from Life Technologies to sequence the libraries with the following kits:
 - Ion PGM Sequencing 200 Kit v2 (Life Technologies, cat. #4482006)
 - Ion 314 Chip Kit v2 for up to $4.0\text{--}5.5 \times 10^5$ reads per sequencing reaction (Life Technologies, cat. #4482261)
 - Ion 318 Chip Kit v2 for up to $4.0\text{--}5.5 \times 10^6$ reads per sequencing reaction (Life Technologies, cat. #4484355)
- Primers with Ion Torrent adaptors (underlined)
 - o A miR30 for: CCATCTCATCCCTGCGTGTCTCCGACTCAG
CTCGAGAAGGTATATTGCT
 - o Loop rev: CCTCTCTATGGGCAGTCGGTGAT TACATCTGTGGCTTCACTA
- Phusion High-Fidelity DNA Polymerase (2 U/ μ L) (Thermo Scientific, cat. #F-530)
- Deoxynucleotide (dNTP) Solution Mix (Promega, cat. #U1330)
- Agencourt AMPure XP – PCR Purification (Beckman Coulter, cat. #A63881)
- DynaMag-Spin/2 Magnet (Life Technologies, cat. #12320D/12321D)
- Ethanol (Sigma-Aldrich, cat. # E7023)
- 1.7 mL Posi-Click Eppendorf Tube (Denville, cat. #C2170)
- PCR Tubes, 120 strips with individual lids (Denville, cat. #C18098-3)
- Nuclease-free water (Qiagen, cat. #129115)

Library preparation

1. Determine the DNA concentration of the isolated genomic DNA with a Qubit Fluorometer using the Qubit dsDNA HS Assay Kit.
2. All shRNAmirs in the library were amplified with three individual PCR reactions containing 100 ng of genomic DNA using the following recipe.

5 x Phusion buffer	10 μ L
10 mM dNTPs	1 μ L
10 μ M A miR30 for	$\square\square\square$ μ L
10 μ M loop rev	2.5 μ L
DNA	100 ng
Phusion Hot Start Taq	0.5 μ L

H ₂ O	Up to 33.5 μ L
Total	50 μ L

3. Primer sequences with the Ion Torrent adaptors underlined
 - a. Primer A+miR30: CCATCTCATCCCTGCGTGTCTCCGACTCAG
CTCGAGAAGGTATATTGCT
 - b. PrimerP1+Loop: CCTCTCTATGGGCAGTCGGTGAT TACATCTGTGGCTTCACTA
4. Amplification protocol
 - a. 98°C – 2 min
 - b. 98°C – 30 sec
 - c. 58°C – 30 sec
 - d. 72°C – 20 sec (b->d 26 cycles)
 - e. 72°C – 2 min
5. The PCR reactions are pooled and proceed to the Ampure XP bead cleanup to remove traces of primer dimers.
6. Keep 5 μ L of the pooled PCR reaction for step 23.
7. Combine PCR reactions in a lo-bind eppendorf tube for 200 μ L final volume (make up volume to 200 μ L with nuclease-free H₂O).
8. Add 260 μ L Ampure beads per sample.
9. Mix by vortexing carefully.
10. Incubate samples $\geq$ 5 min (important – binding to beads is time dependent)
11. Place sample on magnet and wait $\geq$ 5 min. Aspirate supernatant and keep as backup.
12. Wash beads gently twice with 5 volumes of 70% ethanol, while beads stay on magnet.
13. Remove tubes and quick spin to collect residual ethanol.
14. Replace in magnet and aspirate remaining ethanol.
15. Allow samples to dry until beads appear to crack.
16. Elute DNA by adding 20 μ L of Nuclease-free H₂O.
17. Mix by pipetting samples 10 times and incubating at RT for 5 min.
18. Place samples on magnet for 2 min and collect supernatant containing purified DNA.
19. Adjust eluate to 50 μ L final with nuclease-free H₂O
20. Repeat the entire procedure (steps 8-18, then skip to 21) using 80 μ L Ampure beads.
21. Elute the DNA in 20 μ L of nuclease free H₂O.
22. Determine the DNA concentration of the cleaned PCR product as in step 1.
23. Verify that the beads have removed the primer dimers by quantifying the pre-cleaned and cleaned library DNA with the Agilent 2100 Bioanalyzer and the High Sensitivity DNA analysis kit.

PGM sequencing procedure

24. The cleaned library samples are sequenced using the Ion Torrent technology from Life Technologies following the manufacturer's instructions.
25. Dilute the library and use it as a template to prepare template-positive Ion Sphere Particles (ISPs) with the Ion One Touch 200 Template Kit v2 and the Ion One Touch instrument.
26. Perform a quality control test on the template-positive ISPs using the Ion Sphere Quality Control kit and a Qubit Fluorometer. Proceed to enrichment if 10-30% of the ISPs are template-positive.
27. Proceed to the enrichment of the template-positive ISPs with the Ion One Touch 200 Template kit v2 and the Ion One Touch ES.

28. Perform a quality control on the enriched template-positive ISPs using the Ion Sphere Quality Control kit and a Qubit Fluorometer. Proceed to sequencing if more than 50% of the ISPs are template-positive.
29. The enriched template-positive ISPs are loaded on Ion 314 Chip and sequenced with the Ion PGM 200 Sequencing Kit and the Personal Genome Machine System. $2.5\text{-}6 \times 10^6$ reads are routinely obtained with the Ion 314 Chip.

Analysis of sequencing reads from PGM (CD8)

30. Prepare the reference library for analysis by adding ACAGTGAGCG and TAGTGAAGCCACAGATGTA to the 5' and 3' of shRNA sense strand, respectively, of each clone of the 96 constructs used in the experiment.
31. Align the sequencing data with the reference library using BLAST with default mismatch settings.
32. The number of reads for each shRNA_{mir} in the input, effector and memory precursor samples is used in the analysis.
 - a. shRNA_{mir} with read counts lower than 50 in the input were excluded from the analysis.
 - b. shRNA_{mir} with read counts lower than 50 in the effector or memory precursor samples were considered for the analysis if their number of reads was higher than 50 in the input.
 - c. shRNA_{mir} not detected in the effector or memory precursor samples were considered as depleted and a value of 1 was assigned to these shRNA_{mir}.
33. Normalize the number of reads of every shRNA_{mir} in the effector and memory precursor samples were normalized to their respective number of reads in the input sample.
34. The relative enrichment or depletion of shRNA_{mir} from each population is determined by dividing the normalized reads in the memory precursor sample by the normalized reads in the effector sample to obtain the memory precursor/effector ratios for every shRNA_{mir} in the library.
35. Calculate the $\log_2$ of these ratios.
36. Calculate the Z-score values for every shRNA_{mir} with the following formula:

$$Z(\text{shRNA}_{\text{mir}} A) = \frac{\text{Log}_2 \text{ ratio of shRNA}_{\text{mir}} A - \text{mean of log}_2 \text{ ratios of ctrl shRNA}_{\text{mir}}}{\text{standard deviation of ctrl shRNA}_{\text{mir}}}$$
 - a. The Z-score are calculated using control shRNA_{mir} because several shRNA_{mir} in the library were expected to have an effect and therefore the standard deviation would be large and Z-score values small.
 - b. Negative control shRNA_{mir} are specific for genes not expressed in CD8 and therefore are expected to have no effect on the differentiation of effector and memory precursor cells. These shRNA_{mir} were equally distributed in both populations as shown by their clustering the center of the $\log_2$ and Z-score distributions.
 - c. List of negative control shRNA_{mir} used in the CD8 T cell analysis with the current library: *Cd4*, *Cd14*, *Cd19*, *Ms4a1* (CD20).
 - d. All shRNA_{mir} can be used to calculate Z-score values when screening a large shRNA_{mir} library. In this case, it would be expected that the majority of shRNA_{mir} would have no effect on effector and memory precursor differentiation.
37. The Z-score values of every shRNA_{mir} can be plotted.
 - a. A Z-score of $\geq |3|$ Z-score for two or more shRNA_{mir}s against a gene of interest is the primary criterion for further consideration.
38. Another method of visualizing the data takes into account the effects shRNA_{mir} have on cell accumulation (possible effects on survival or proliferation).
 - a. For this analysis the effector/input and memory precursor/input ratios are obtained by dividing the normalized reads in the effector or memory precursor samples by the reads in the input sample for every shRNA_{mir} in the library.

- b. The $\log_2$ values of the effector/input and memory precursor/input ratios are calculated.
- c. Plot the $\log_2$ values of the effector/input ratios against those of the memory precursor/input ratios.

Detailed CD4 deep sequencing and analysis

Materials

- Qubit dsDNA HS Assay Kit (Life Technologies, cat. # Q32854)
- Qubit Fluorometer (Life Technologies)
- PCR machine
- Agilent 2100 Bioanalyzer (Agilent Technologies)
- High Sensitivity DNA analysis kit (Agilent Technologies, cat. #5067-4626)
- Deep sequencing machine
 - o We use the Ion OneTouch 2 System from Life Technologies to prepare the sequencing libraries with the following kits:
 - Ion OneTouch 200 Template Kit v2 DL (Life Technologies, cat. #4480285)
 - Ion Sphere Quality Control Kit (Life Technologies, cat. #4468656)
 - o We use the Ion PGM System from Life Technologies to sequence the libraries with the following kits:
 - Ion PGM Sequencing 200 Kit v2 (Life Technologies, cat. #4482006)
 - Ion 314 Chip Kit v2 for up to $4.0\text{--}5.5 \times 10^5$ reads per sequencing reaction (Life Technologies, cat. #4482261)
 - Ion 318 Chip Kit v2 for up to $4.0\text{--}5.5 \times 10^6$ reads per sequencing reaction (Life Technologies, cat. #4484355)
- Primers with Ion Torrent adaptors (underlined)
 - o A miR30 for: CCATCTCATCCCTGCGTGTCTCCGACTCAG
CTCGAGAAGGTATATTGCT
 - o Loop rev: CCTCTCTATGGGCAGTCGGTGAT TACATCTGTGGCTTCACTA
- Phusion High-Fidelity DNA Polymerase (2 U/ μ L) (Thermo Scientific, cat. #F-530)
- Deoxynucleotide (dNTP) Solution Mix (New England BioLabs, cat. #N0447)
- Agencourt AMPure XP – PCR Purification (Beckman Coulter, cat. #A63881)
- DynaMag-Spin/2 Magnet (Life Technologies, cat. #12320D/12321D)
- Ethanol (Sigma-Aldrich, cat. # E7023)
- 1.7 mL Posi-Click Eppendorf Tube (Denville, cat. #C2170)
- PCR Tubes, 120 strips with individual lids (Denville, cat. #C18098-3)
- Nuclease-free water (Qiagen, cat. #129115)

Library preparation

1. Determine the DNA concentration of the isolated genomic DNA with a Qubit Fluorometer using the Qubit dsDNA HS Assay Kit as per manufacturer's instructions.
2. All shRNAmirs in the library were amplified with three individual PCR reactions containing 100 ng of genomic DNA using the following recipe.

5 x Phusion buffer	10 μ L
10 mM dNTPs	1 μ L
10 μ M A miR30 for	$\square\square\square$ μ L
10 μ M loop rev	2.5 μ L
DNA	100 ng
Phusion Hot Start Taq	0.5 μ L
H ₂ O	Up to 33.5 $\square$ μ L
Total	50 $\square$ μ L

3. Amplification protocol
 - a. 98°C – 2 min
 - b. 98°C – 30 sec

- c. 58°C – 30 sec
 - d. 72°C – 20 sec -> b-d: 26 cycles
 - e. 72°C – 2 min
4. The PCR reactions are pooled and 15 µL is loaded on a 1.5% agarose gel to verify the efficiency of the PCR reaction.
 - a. The amplified library should be visible as a band of 127 bp.
 5. Proceed to the cleanup of PCR reaction to remove traces of primer dimers.
 6. Keep 5 µL of the pooled PCR reaction for step 23.
 7. Combine PCR reactions in a 1.7 mL Eppendorf tube for 200 µL final volume (make up volume to 200 µL with nuclease-free H₂O).
 8. Add 260 µL of AMPure XP beads per sample.
 9. Mix by vortexing carefully.
 10. Incubate samples ≥ 5 min (important – binding to beads is time dependent)
 11. Place sample on magnet and wait ≥ 5 min. Aspirate supernatant and keep as backup.
 12. Wash beads gently twice with 1 mL of 70% ethanol, while beads stay on magnet.
 13. Remove tubes and quickly spin to collect residual ethanol.
 14. Replace in magnet and aspirate remaining ethanol.
 15. Allow samples to dry until beads appear to crack.
 16. Elute DNA by adding 20 µL of nuclease-free H₂O.
 17. Mix by pipetting samples 10 times and incubating at RT for 5 min.
 18. Place samples on magnet for 2 min and collect supernatant containing purified DNA.
 19. Adjust eluate to 50 µL final with nuclease-free H₂O
 20. Repeat the entire procedure (steps 46-56, then skip to step 59) using 80 µL of AMPure XP beads.
 21. Elute the DNA in 20 µL of nuclease free H₂O.
 22. Determine the DNA concentration of the cleaned PCR product as in step 39.
 23. Verify that the beads have removed the primer dimers by quantifying the pre-cleaned (kept at step 6) and cleaned library DNA (step 21) with the Agilent 2100 Bioanalyzer and the High Sensitivity DNA analysis kit as per manufacturer's instructions.
 - a. The result obtained with the Agilent Bioanalyzer should show the disappearance of the primer dimers after the cleanup of the PCR reaction with the AMPure beads XP.
 - b. Only once the primer dimers have been successfully removed from the amplified library that the library can be sequenced.

PGM sequencing procedure

24. The cleaned library samples are sequenced using the Ion Torrent technology from Life Technologies following the manufacturer's instructions.
25. Dilute the library and use it as a template to prepare template-positive Ion Sphere Particles (ISPs) with the Ion One Touch 200 Template Kit v2 DL and the Ion OneTouch 2 System following manufacturer's instructions.
 - a. The dilution of the library is critical for an efficient sequencing reaction. The dilution factor to obtain a library that will be efficiently sequenced needs to be experimentally determined. We typically need to dilute the library by a factor of 50 to 267.
 - b. The reaction can be started at the end of the day and the template-positive ISPs can be recovered the following morning.
26. Perform a quality control test on the template-positive ISPs using the Ion Sphere Quality Control kit and a Qubit Fluorometer following manufacturer's instructions.
 - a. Proceed to enrichment if 10-30% of the ISPs are template-positive.
 - b. If less than 10% of the ISPs are template-positive, the sequencing reaction will not generate enough reads for analysis.
 - c. If more than 30% of the ISPs are template-positive, some ISPs will be multi-templated and will generate mixed reads.

27. Enrich the template-positive ISPs with the Ion One Touch 200 Template kit v2 DL and the Ion OneTouch System following manufacturer's instructions.
28. Perform a quality control on the enriched template-positive ISPs using the Ion Sphere Quality Control kit and a Qubit Fluorometer following manufacturer's instructions.
 - a. Proceed to sequencing if more than 50% of the ISPs are template-positive.
29. The enriched template-positive ISPs are loaded on Ion 314 or Ion 318 Chip and sequenced with the Ion PGM 200 Sequencing Kit and the Personal Genome Machine System following manufacturer's instructions.
 - a. 250,000-500,000 reads are routinely obtained with the Ion 314 Chip.
 - b. The higher density Ion 318 Chip can yield up to 10 times more reads than the Ion 314 Chip.
30. Prepare the reference library for analysis by adding ACAGTGAGCG and TAGTGAAGCCACAGATGTA to the 5' and 3' of shRNA sense strand, respectively, of each shRNAmir sequences used in the experiment.
31. Align the sequencing data with the reference library using BLAST with default mismatch settings.
32. The number of reads for each shRNAmir in the input, Tfh and Th1 samples is used in the analysis.
 - a. shRNAmir with read counts lower than 50 in the input were excluded from the analysis.
 - b. shRNAmir with read counts lower than 50 in the Tfh or Th1 samples were considered for the analysis if their number of reads was higher than 50 in the input.
 - c. shRNAmir not detected in the Tfh or Th1 samples were considered as depleted and a value of 1 was assigned to these shRNAmir.
33. Normalize the number of reads of every shRNAmir in the Tfh and Th1 samples to their respective number of reads in the input sample.
34. The relative enrichment or depletion of shRNAmir from each population is determined by dividing the normalized reads in the Tfh sample by the normalized reads in the Th1 sample to obtain the Tfh/Th1 ratios for every shRNAmir in the library.
35. Calculate the $\log_2$ of these ratios.
36. Calculate the Z-score values for every shRNAmir with the following formula:

$$Z(\text{shRNAmir } A) = \frac{\text{Log2 ratio of shRNAmir } A - \text{mean of log2 ratios of ctrl shRNAmir}}{\text{standard deviation of ctrl shRNAmir}}$$
 - a. The Z-score are calculated using control shRNAmir because several shRNAmir in the library were expected to have an effect and therefore the standard deviation would be large and Z-score values small.
 - b. Negative control shRNAmir are targeting genes not expressed in CD4 and therefore are expected to have no effect on the differentiation of Tfh and Th1 cells. These shRNAmir were equal ly distributed in both populations as shown by their clustering the center of the $\log_2$ and Z-score distributions.
 - c. List of negative control shRNAmir used in the CD4 T cell analysis with the current library: *Cd14*, *Cd19*, *Cd22*, *Ms4a1* (CD20), *Cd8* and *Smarca1*.
 - d. All shRNAmir can be used to calculate Z-score values when screening a large shRNAmir library. In this case, it would be expected that the majority of shRNAmir would have no effect on Tfh and Th1 differentiation.
37. The Z-score values of every shRNAmir can be plotted.
 - a. A Z-score of $\geq |3|$ Z-score for at least one shRNAmir against a gene of interest is the primary criterion for further consideration.
 - b. If shRNAmirs against a gene are giving conflicting outcomes (one shRNAmir with a Tfh bias and a Th1 bias for a second shRNAmir), the gene was discarded.
38. Another method of visualizing the data takes into account the effects shRNAmir have on cell accumulation (possible effects on survival or proliferation).
 - a. For this analysis the Tfh/input and Th1/input ratios are obtained by dividing the normalized reads in the Tfh or Th1 samples by the reads in the input sample for every shRNAmir in the library.
 - b. The $\log_2$ values of the Tfh/input and Th1/input ratios are calculated.
 - c. Plot the $\log_2$ values of the Tfh/input ratios against those of the Th1/input ratios.

Detailed shRNAmir sub-cloning protocol

Materials

- GIPZ mouse shRNA whole-genome library (Thermo Scientific)
- Whatman 2 ml deepwell block (Fisher, cat. #05-714-034)
- LB Broth LENNOX (VWR, cat. # EM1.00547.0500)
- Ampicillin (Fisher, cat. # BP1760-25)
- Glycerol (Fisher, cat. # BP229-4)
- Aeraseal (Fisher, cat. # 50212463)
- GenElute™ HP 96 Well Plasmid Miniprep Kit (Sigma Aldrich, cat. # NA9604)
- XhoI (New England Biolabs, cat. # R0146)
- EcoRI (New England Biolabs, cat. # R0101)
- Hot Start II High-Fidelity DNA polymerase (Thermo Scientific, cat. # F-549)
- QIAquick Gel Extraction Kit (Qiagen, cat. # 28706)
- Rapid Ligation Kit (Thermo Scientific, cat. # K1423)
- XL10-Gold chemically competent cells (Agilent Technologies, cat. # 200315)

Culture preparation

1. Transfer 5 µL of glycerol stock of GIPZ clones into a Whatman 2 mL deepwell block containing 1.2 mL of 2X LB with 100 µg/mL Ampicillin.
2. Culture for ~24 hours at 37°C with vigorous shaking (~300 rpm)
3. Transfer 5 µL of the starter cell culture into a 2 mL square well block containing 1.2 mL of 2X LB with 100 µg/mL Ampicillin.
4. Culture for 16-17 hours at 37°C with vigorous shaking (~300 rpm)

96 well DNA miniprep

5. DNA is purified with GenElute™ HP 96 Well Plasmid Miniprep Kit.
6. Pellet the 96 deep well culture plate containing the overnight culture by centrifuging at 1800 x g for 10 minutes.
7. During centrifugation, assemble the manifold by placing an empty deep well collection plate in the manifold base. Place the manifold collar over the deep well collection plate so it fits securely on top of the manifold base. Place the filter plate on top of the manifold collar.
8. Following centrifugation of the deep well culture plate, pour off the supernatant and blot the deep well culture plate upside down on an absorbent pad.
9. Using a multichannel pipette, add 200 µL of Resuspension Solution to the pelleted cells and resuspend by pipetting up and down.
10. Add 200 µL of Lysis Solution and pipette up and down 5-6 times to mix.
11. Allow the lysate to incubate at room temperature for 3 to 5 minutes; do not allow the lysis to exceed 5 minutes.
12. Add 200 µL of Neutralization Solution and pipette up and down 5-6 times to mix.
13. Add 200 µL of Binding Solution and pipette up and down 2-3 times to mix.
14. Transfer the lysate mix from the culture plate to the filter plate that is located on top of the manifold assembly and let sit for 5 minutes.
15. Apply vacuum. Allow the lysate to filter completely through the filter plate and into the deep well collection plate. Depending on the cell culture density this may take a few minutes. Once the lysate is completely filtered, turn off and release vacuum.
16. Discard filter plate and prepare the manifold by removing the manifold collar. Remove the deep well collection plate containing the filtered lysate from the manifold base and set aside. Place the waste tray in the manifold base. Place the manifold collar over the waste tray so it fits securely on top of the manifold base. Place the binding plate on top of the manifold collar.
17. Add 600 µL of Column Preparation Solution and apply vacuum until all the solution passes through the binding plate. Turn off vacuum.
18. Transfer the filtered lysate from the deep well collection plate to the binding plate.

19. Apply vacuum so that the solution slowly passes through the binding plate. Turn off vacuum.
20. Add 600 μ L of Wash Solution 1 and apply vacuum until all the solution passes through the binding plate. Turn off vacuum.
21. Add 600 μ L of Wash Solution 2 and apply vacuum until all the solution passes through the binding plate. Turn off vacuum.
22. Remove the binding plate from the vacuum manifold and thoroughly blot the plate on an absorbent pad, making sure all residual Wash Solution 2 is removed.
23. Place the binding plate back on the vacuum manifold, apply maximum vacuum and dry the binding plate for a minimum of 10 minutes.
24. Once the plate is dry, place the binding plate on top of the elution plate.
25. Add 100 μ L of Elution Solution directly onto the membrane for each well of the binding plate and centrifuge the plate at 1800 x g for 5 minutes.

shRNAmir inserts preparation

26. shRNAmir inserts were prepared using two approaches:

Direct approach:

- a. Pool the purified plasmid DNA from the original GIPZ clones.
- b. Cleave the pool DNA with XhoI and EcoRI to release all 96 inserts simultaneously.
- c. Purify the inserts after restriction enzyme digestion using QIAquick gel extraction kit.

PCR approach:

- a. Amplify each individual shRNAmir insert in 20 μ L reactions containing 5 ng of plasmid DNA using the following recipe.

5 x Phusion buffer	4 μ L
10 mM dNTPs	0.4 μ L
10 μ M A XhoI-5'mir30	1 μ L
10 μ M EcoRI-3'mir30	1 μ L
DNA	5 ng
Phusion Hot Start II High-Fidelity DNA polymerase	0.2 μ L
H ₂ O	Up to 13.4 μ L
Total	20 μ L

- b. Primer sequences with the restriction enzymes sequences underlined:

XhoI-5'mir30:

CAGAAGGCTCGAGAAAGGTATATTGCTGTTGACAGTGAGCG

EcoRI-3'mir30:

CTAAAGTAGCCCCTTGAATTCGAGGCAGTAGGCA

- c. Amplification protocol.

- a. 94°C – 5 min
- b. 94°C – 30 sec
- c. 54°C – 30 sec
- d. 75°C – 30 sec (b->d 16 cycles)
- e. 75°C – 2 min
- d. Confirm the amplification of each shRNAmir with agarose gel electrophoresis.
- e. Pool the PCR products and digest with DpnI to destroy remaining template plasmid DNA for 1 hour at 37°C.
- f. Digest the pooled PCR products with EcoRI and XhoI to generate sticky ends.
- g. Purify the inserts after restriction enzyme digestion using QIAquick gel extraction kit.

Ligation and transformation

27. Ligate inserts prepared with each method separately into 100 ng of pLMPd-Amt in a 2.5:1 molar ratio.
28. Transform 10% of each ligation into 50 μ L of XL10-Gold chemically competent cells.
29. Pick 384 colonies from each transformation.
30. Culture in 96-well plates containing 200 μ L of LB medium with 8% of glycerol and 50 μ g/mL Ampicillin for 24 hour at 37°C without shaking.
31. Mix the culture and freeze in -80°C.

DNA sequencing and analysis

32. Send the frozen culture to Beckman Genomics for DNA sequencing.
33. Align all reads to a reference file from the 96 shRNA sequences of the original GIPZ library using the BLAST tool.

Supplemental References

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